

FastAPI Reference

FastAPI class

Here's the reference information for the `FastAPI` class, with all its parameters, attributes and methods.

You can import the `FastAPI` class directly from `fastapi`:

```
from fastapi import FastAPI
```

`class` `fastapi.FastAPI`

```
FastAPI(
    *,
    debug=False,
    routes=None,
    title="FastAPI",
    summary=None,
    description="",
    version="0.1.0",
    openapi_url="/openapi.json",
    openapi_tags=None,
    servers=None,
    dependencies=None,
    default_response_class=Default(JSONResponse),
    redirect_slashes=True,
    docs_url="/docs",
    redoc_url="/redoc",
    swagger_ui_oauth2_redirect_url="/docs/oauth2-redirect",
    swagger_ui_init_oauth=None,
    middleware=None,
    exception_handlers=None,
    on_startup=None,
    on_shutdown=None,
    lifespan=None,
    terms_of_service=None,
    contact=None,
    license_info=None,
    openapi_prefix="",
    root_path="",
    root_path_in_servers=True,
    responses=None,
    callbacks=None,
    webhooks=None,
    deprecated=None,
    include_in_schema=True,
    swagger_ui_parameters=None,
    generate_unique_id_function=Default(generate_unique_id),
    separate_input_output_schemas=True,
    openapi_external_docs=None,
    strict_content_type=True,
    **extra
)
```

Bases: `Starlette`

`FastAPI` app class, the main entrypoint to use FastAPI.

Read more in the [FastAPI docs for First Steps](#).

Example

```
from fastapi import FastAPI

app = FastAPI()
```

PARAMETER	DESCRIPTION
debug	Boolean indicating if debug tracebacks should be returned on server errors. Read more in the Starlette docs for Applications. TYPE: <code>bool</code> DEFAULT: <code>False</code>
routes	Note: you probably shouldn't use this parameter, it is inherited from Starlette and supported for compatibility. A list of routes to serve incoming HTTP and WebSocket requests. TYPE: <code>list[BaseRoute]</code> <code>None</code> DEFAULT: <code>None</code>
title	The title of the API. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more in the FastAPI docs for Metadata and Docs URLs. Example <pre>from fastapi import FastAPI app = FastAPI(title="ChimichangApp")</pre> TYPE: <code>str</code> DEFAULT: <code>'FastAPI'</code>
summary	A short summary of the API. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more in the FastAPI docs for Metadata and Docs URLs. Example <pre>from fastapi import FastAPI app = FastAPI(summary="Deadpond's favorite app. Nuff said.")</pre> TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
description	A description of the API. Supports Markdown (using CommonMark syntax). It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more in the FastAPI docs for Metadata and Docs URLs. Example <pre>from fastapi import FastAPI app = FastAPI(description=""" ChimichangApp API helps you do awesome stuff. """</pre>

PARAMETER	DESCRIPTION
	<pre>## Items You can **read items**. ## Users You will be able to: * **Create users** (_not implemented_). * **Read users** (_not implemented_). """)</pre> TYPE: str DEFAULT: '0.1.0'
version	The version of the API. Note This is the version of your application, not the version of the OpenAPI specification nor the version of FastAPI being used. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more in the FastAPI docs for Metadata and Docs URLs. Example <pre>from fastapi import FastAPI app = FastAPI(version="0.0.1")</pre> TYPE: str DEFAULT: '0.1.0'
openapi_url	The URL where the OpenAPI schema will be served from. If you set it to <code>None</code>, no OpenAPI schema will be served publicly, and the default automatic endpoints <code>/docs</code> and <code>/redoc</code> will also be disabled. Read more in the FastAPI docs for Metadata and Docs URLs. Example <pre>from fastapi import FastAPI app = FastAPI(openapi_url="/api/v1/openapi.json")</pre> TYPE: str None DEFAULT: '/openapi.json'
openapi_tags	A list of tags used by OpenAPI, these are the same <code>tags</code> you can set in the <i>path operations</i>, like: <code>@app.get("/users/", tags=["users"])</code> <code>@app.get("/items/", tags=["items"])</code> The order of the tags can be used to specify the order shown in tools like Swagger UI, used in the automatic path <code>/docs</code>. It's not required to specify all the tags used. The tags that are not declared MAY be organized randomly or based on the tools' logic. Each tag name in the list MUST be unique. The value of each item is a <code>dict</code> containing:

PARAMETER	DESCRIPTION <code>name</code>: The name of the tag. <code>description</code>: A short description of the tag. CommonMark syntax  MAY be used for rich text representation. <code>externalDocs</code>: Additional external documentation for this tag. If provided, it would contain a <code>dict</code> with: <code>description</code>: A short description of the target documentation. CommonMark syntax  MAY be used for rich text representation. <code>url</code>: The URL for the target documentation. Value MUST be in the form of a URL. Read more in the FastAPI docs for Metadata and Docs URLs . Example <pre>from fastapi import FastAPI tags_metadata = [{ "name": "users", "description": "Operations with users. The login logic is also here.", }, { "name": "items", "description": "Manage items. So <i>fancy</i> they have their own docs.", "externalDocs": { "description": "Items external docs", "url": "https://fastapi.tiangolo.com/", }, },] app = FastAPI(openapi_tags=tags_metadata)</pre> TYPE: <code>list[dict[str, Any]] None</code> DEFAULT: <code>None</code>
<code>servers</code>	A <code>list</code> of <code>dict</code>s with connectivity information to a target server. You would use it, for example, if your application is served from different domains and you want to use the same Swagger UI in the browser to interact with each of them (instead of having multiple browser tabs open). Or if you want to leave fixed the possible URLs. If the <code>servers</code> <code>list</code> is not provided, or is an empty <code>list</code>, the <code>servers</code> property in the generated OpenAPI will be: - a <code>dict</code> with a <code>url</code> value of the application's mounting point (<code>root_path</code>) if it's different from <code>/</code>. - otherwise, the <code>servers</code> property will be omitted from the OpenAPI schema. Each item in the <code>list</code> is a <code>dict</code> containing: <code>url</code>: A URL to the target host. This URL supports Server Variables and MAY be relative, to indicate that the host location is relative to the location where the OpenAPI document is being served. Variable substitutions will be made when a variable is named in <code>{ brackets }</code>.

PARAMETER	DESCRIPTION
	<code>description</code>: An optional string describing the host designated by the URL. CommonMark syntax MAY be used for rich text representation. <code>variables</code>: A <code>dict</code> between a variable name and its value. The value is used for substitution in the server's URL template. Read more in the FastAPI docs for Behind a Proxy. Example <pre>from fastapi import FastAPI app = FastAPI(servers=[{"url": "https://stag.example.com", "description": "Staging environment"}, {"url": "https://prod.example.com", "description": "Production environment"},])</pre> TYPE: <code>list[dict[str, str Any]] None</code> DEFAULT: <code>None</code>
<code>dependencies</code>	A list of global dependencies, they will be applied to each <i>path operation</i>, including in sub-routers. Read more about it in the FastAPI docs for Global Dependencies. Example <pre>from fastapi import Depends, FastAPI from .dependencies import func_dep_1, func_dep_2 app = FastAPI(dependencies=[Depends(func_dep_1), Depends(func_dep_2)])</pre> TYPE: <code>Sequence[Depends] None</code> DEFAULT: <code>None</code>
<code>default_response_class</code>	The default response class to be used. Read more in the FastAPI docs for Custom Response - HTML, Stream, File, others. Example <pre>from fastapi import FastAPI from fastapi.responses import ORJSONResponse app = FastAPI(default_response_class=ORJSONResponse)</pre> TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>redirect_slashes</code>	Whether to detect and redirect slashes in URLs when the client doesn't use the same format. Example <pre>from fastapi import FastAPI app = FastAPI(redirect_slashes=True) # the default @app.get("/items/")</pre>

PARAMETER	DESCRIPTION
	<pre>async def read_items(): return [{"item_id": "Foo"}]</pre> With this app, if a client goes to <code>/items</code> (without a trailing slash), they will be automatically redirected with an HTTP status code of <code>307</code> to <code>/items/</code>. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>docs_url</code>	The path to the automatic interactive API documentation. It is handled in the browser by Swagger UI. The default URL is <code>/docs</code>. You can disable it by setting it to <code>None</code>. If <code>openapi_url</code> is set to <code>None</code>, this will be automatically disabled. Read more in the FastAPI docs for Metadata and Docs URLs. Example <pre>from fastapi import FastAPI app = FastAPI(docs_url="/documentation", redoc_url=None)</pre> TYPE: <code>str None</code> DEFAULT: <code>'/docs'</code>
<code>redoc_url</code>	The path to the alternative automatic interactive API documentation provided by ReDoc. The default URL is <code>/redoc</code>. You can disable it by setting it to <code>None</code>. If <code>openapi_url</code> is set to <code>None</code>, this will be automatically disabled. Read more in the FastAPI docs for Metadata and Docs URLs. Example <pre>from fastapi import FastAPI app = FastAPI(docs_url="/documentation", redoc_url="redocumentation")</pre> TYPE: <code>str None</code> DEFAULT: <code>'/redoc'</code>
<code>swagger_ui_oauth2_redirect_url</code>	The OAuth 2 redirect endpoint for the Swagger UI. By default it is <code>/docs/oauth2-redirect</code>. This is only used if you use OAuth 2 (with the "Authorize" button) with Swagger UI. TYPE: <code>str None</code> DEFAULT: <code>'/docs/oauth2-redirect'</code>
<code>swagger_ui_init_oauth</code>	OAuth 2 configuration for the Swagger UI, by default shown at <code>/docs</code>. Read more about the available configuration options in the Swagger UI docs. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>middleware</code>	List of middleware to be added when creating the application. In FastAPI you would normally do this with <code>app.add_middleware()</code> instead.

PARAMETER	DESCRIPTION
	Read more in the FastAPI docs for Middleware. TYPE: <code>Sequence[Middleware] None</code> DEFAULT: <code>None</code>
<code>exception_handlers</code>	A dictionary with handlers for exceptions. In FastAPI, you would normally use the decorator <code>@app.exception_handler()</code>. Read more in the FastAPI docs for Handling Errors. TYPE: <code>dict[int type[Exception], Callable[[Request, Any], Coroutine[Any, Any, Response]] None</code> DEFAULT: <code>None</code>
<code>on_startup</code>	A list of startup event handler functions. You should instead use the <code>lifespan</code> handlers. Read more in the FastAPI docs for lifespan. TYPE: <code>Sequence[Callable[[], Any]] None</code> DEFAULT: <code>None</code>
<code>on_shutdown</code>	A list of shutdown event handler functions. You should instead use the <code>lifespan</code> handlers. Read more in the FastAPI docs for lifespan. TYPE: <code>Sequence[Callable[[], Any]] None</code> DEFAULT: <code>None</code>
<code>lifespan</code>	A <code>Lifespan</code> context manager handler. This replaces <code>startup</code> and <code>shutdown</code> functions with a single context manager. Read more in the FastAPI docs for lifespan. TYPE: <code>Lifespan[AppType] None</code> DEFAULT: <code>None</code>
<code>terms_of_service</code>	A URL to the Terms of Service for your API. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more at the FastAPI docs for Metadata and Docs URLs. Example <pre>app = FastAPI(terms_of_service="http://example.com/terms/")</pre> TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>contact</code>	A dictionary with the contact information for the exposed API. It can contain several fields. <code>name: (str)</code> The name of the contact person/organization. <code>url: (str)</code> A URL pointing to the contact information. MUST be in the format of a URL.

PARAMETER	DESCRIPTION
	<code>email: (str)</code> The email address of the contact person/organization. MUST be in the format of an email address. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more at the FastAPI docs for Metadata and Docs URLs. Example <pre>app = FastAPI(contact={ "name": "Deadpoolio the Amazing", "url": "http://x-force.example.com/contact/", "email": "dp@x-force.example.com", })</pre> TYPE: <code>dict[str, str Any] None</code> DEFAULT: <code>None</code>
<code>license_info</code>	A dictionary with the license information for the exposed API. It can contain several fields. <code>name: (str)</code> REQUIRED (if a <code>license_info</code> is set). The license name used for the API. <code>identifier: (str)</code> An SPDX license expression for the API. The <code>identifier</code> field is mutually exclusive of the <code>url</code> field. Available since OpenAPI 3.1.0, FastAPI 0.99.0. <code>url: (str)</code> A URL to the license used for the API. This MUST be the format of a URL. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more at the FastAPI docs for Metadata and Docs URLs. Example <pre>app = FastAPI(license_info={ "name": "Apache 2.0", "url": "https://www.apache.org/licenses/LICENSE-2.0.html", })</pre> TYPE: <code>dict[str, str Any] None</code> DEFAULT: <code>None</code>
<code>openapi_prefix</code>	A URL prefix for the OpenAPI URL. TYPE: <code>str</code> DEFAULT: <code>''</code>

PARAMETER	DESCRIPTION
<code>root_path</code>	A path prefix handled by a proxy that is not seen by the application but is seen by external clients, which affects things like Swagger UI. Read more about it at the FastAPI docs for Behind a Proxy. Example <pre>from fastapi import FastAPI app = FastAPI(root_path="/api/v1")</pre> TYPE: <code>str</code> DEFAULT: <code>''</code>
<code>root_path_in_servers</code>	To disable automatically generating the URLs in the <code>servers</code> field in the autogenerated OpenAPI using the <code>root_path</code>. Read more about it in the FastAPI docs for Behind a Proxy. Example <pre>from fastapi import FastAPI app = FastAPI(root_path_in_servers=False)</pre> TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>responses</code>	Additional responses to be shown in OpenAPI. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Additional Responses in OpenAPI. And in the FastAPI docs for Bigger Applications. TYPE: <code>dict[int str, dict[str, Any]] None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	OpenAPI callbacks that should apply to all <i>path operations</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>webhooks</code>	Add OpenAPI webhooks. This is similar to <code>callbacks</code> but it doesn't depend on specific <i>path operations</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Note: This is available since OpenAPI 3.1.0, FastAPI 0.99.0. Read more about it in the FastAPI docs for OpenAPI Webhooks. TYPE: <code>APIRouter</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark all <i>path operations</i> as deprecated. You probably don't need it, but it's available. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>).

PARAMETER	DESCRIPTION
	Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>include_in_schema</code>	To include (or not) all the <i>path operations</i> in the generated OpenAPI. You probably don't need it, but it's available. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>swagger_ui_parameters</code>	Parameters to configure Swagger UI, the autogenerated interactive API documentation (by default at <code>/docs</code>). Read more about it in the FastAPI docs about how to Configure Swagger UI. TYPE: <code>dict[str, Any]</code> <code>None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>
<code>separate_input_output_schemas</code>	Whether to generate separate OpenAPI schemas for request body and response body when the results would be more precise. This is particularly useful when automatically generating clients. For example, if you have a model like: <pre>from pydantic import BaseModel class Item(BaseModel): name: str tags: list[str] = []</pre> When <code>Item</code> is used for input, a request body, <code>tags</code> is not required, the client doesn't have to provide it. But when using <code>Item</code> for output, for a response body, <code>tags</code> is always available because it has a default value, even if it's just an empty list. So, the client should be able to always expect it. In this case, there would be two different schemas, one for input and another one for output. Read more about it in the FastAPI docs about how to separate schemas for input and output. TYPE: <code>bool</code> DEFAULT: <code>True</code>

PARAMETER	DESCRIPTION
<code>openapi_external_docs</code>	This field allows you to provide additional external documentation links. If provided, it must be a dictionary containing: <code>description</code>: A brief description of the external documentation. <code>url</code>: The URL pointing to the external documentation. The value MUST be a valid URL format. Example: <pre>from fastapi import FastAPI external_docs = { "description": "Detailed API Reference", "url": "https://example.com/api-docs", } app = FastAPI(openapi_external_docs=external_docs)</pre> TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>strict_content_type</code>	Enable strict checking for request Content-Type headers. When <code>True</code> (the default), requests with a body that do not include a <code>Content-Type</code> header will not be parsed as JSON. This prevents potential cross-site request forgery (CSRF) attacks that exploit the browser's ability to send requests without a Content-Type header, bypassing CORS preflight checks. In particular applicable for apps that need to be run locally (in localhost). When <code>False</code>, requests without a <code>Content-Type</code> header will have their body parsed as JSON, which maintains compatibility with certain clients that don't send <code>Content-Type</code> headers. Read more about it in the FastAPI docs for Strict Content-Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>**extra</code>	Extra keyword arguments to be stored in the app, not used by FastAPI anywhere. TYPE: <code>Any</code> DEFAULT: <code>{}</code>

<> Source code in `fastapi/applications.py`

```

57 def __init__(
58     self: AppType,
59     *,
60     debug: Annotated[
61         bool,
62         Doc(
63             """
64             Boolean indicating if debug tracebacks should be returned on server
65             errors.
66
67             Read more in the
68             [Starlette docs for Applications]
69             (https://www.starlette.dev/applications/#instantiating-the-application).
70             """
71         ),
72     ] = False,
73     routes: Annotated[
74         list[BaseRoute] | None,
75         Doc(
76             """
77             **Note**: you probably shouldn't use this parameter, it is inherited
78             from Starlette and supported for compatibility.
79
80             ---
81
82             A list of routes to serve incoming HTTP and WebSocket requests.
83             """
84         ),
85     deprecated(
86         """
87         You normally wouldn't use this parameter with FastAPI, it is inherited
88         from Starlette and supported for compatibility.
89
90         In FastAPI, you normally would use the *path operation methods*,
91         like `app.get()`, `app.post()`, etc.
92         """
93     ),
94     ] = None,
95     title: Annotated[
96         str,
97         Doc(
98             """
99             The title of the API.
100
101             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
102
103             Read more in the
104             [FastAPI docs for Metadata and Docs URLs]
105             (https://fastapi.tiangolo.com/tutorial/metadata/#metadata-for-api).
106
107             **Example**
108
109             ```python
110             from fastapi import FastAPI
111
112             app = FastAPI(title="ChimichangApp")
113             ```
114             """
115         ),
116     ] = "FastAPI",
117     summary: Annotated[
118         str | None,
119         Doc(
120             """
121             A short summary of the API.
122
123             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
124
125             Read more in the
126             [FastAPI docs for Metadata and Docs URLs]
127             (https://fastapi.tiangolo.com/tutorial/metadata/#metadata-for-api).
128

```

```

129         **Example**
130
131         ```python
132         from fastapi import FastAPI
133
134         app = FastAPI(summary="Deadpond's favorite app. Nuff said.")
135         """
136
137     ),
138     ] = None,
139     description: Annotated[
140         str,
141         Doc(
142             """
143             A description of the API. Supports Markdown (using
144             [CommonMark syntax](https://commonmark.org/)).
145
146             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
147
148             Read more in the
149             [FastAPI docs for Metadata and Docs URLs]
150             (https://fastapi.tiangolo.com/tutorial/metadata/#metadata-for-api).
151         """
152     ),
153     ] = None,
154     """python
155     from fastapi import FastAPI
156
157     app = FastAPI(
158         description="""
159         ChimichangApp API helps you do awesome stuff. 🌮
160
161         ## Items
162
163         You can **read items**.
164
165         ## Users
166
167         You will be able to:
168
169         * **Create users** (_not implemented_).
170         * **Read users** (_not implemented_).
171         """
172     ),
173     """
174
175     """
176
177     ),
178     ] = None,
179     version: Annotated[
180         str,
181         Doc(
182             """
183             The version of the API.
184
185             **Note** This is the version of your application, not the version of
186             the OpenAPI specification nor the version of FastAPI being used.
187
188             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
189
190             Read more in the
191             [FastAPI docs for Metadata and Docs URLs]
192             (https://fastapi.tiangolo.com/tutorial/metadata/#metadata-for-api).
193         """
194     ),
195     """python
196     from fastapi import FastAPI
197
198     app = FastAPI(version="0.0.1")
199     """
200
201     ),
202     ] = "0.1.0",
203     openapi_url: Annotated[
204         str | None,

```

```

205     Doc(
206         """
207         The URL where the OpenAPI schema will be served from.
208
209         If you set it to `None`, no OpenAPI schema will be served publicly, and
210         the default automatic endpoints `/docs` and `/redoc` will also be
211         disabled.
212
213         Read more in the
214         [FastAPI docs for Metadata and Docs URLs]
215         (https://fastapi.tiangolo.com/tutorial/metadata/#openapi-url).
216
217         **Example**
218
219         ```python
220         from fastapi import FastAPI
221
222         app = FastAPI(openapi_url="/api/v1/openapi.json")
223         """
224     ),
225     openapi_tags: Annotated[
226         list[dict[str, Any]] | None,
227         Doc(
228             """
229             A list of tags used by OpenAPI, these are the same `tags` you can set
230             in the *path operations*, like:
231
232             * `@app.get("/users/", tags=["users"])`
233             * `@app.get("/items/", tags=["items"])`
234
235             The order of the tags can be used to specify the order shown in
236             tools like Swagger UI, used in the automatic path `/docs`.
237
238             It's not required to specify all the tags used.
239
240             The tags that are not declared MAY be organized randomly or based
241             on the tools' logic. Each tag name in the list MUST be unique.
242
243             The value of each item is a `dict` containing:
244
245             * `name`: The name of the tag.
246             * `description`: A short description of the tag.
247               [CommonMark syntax](https://commonmark.org/) MAY be used for rich
248               text representation.
249             * `externalDocs`: Additional external documentation for this tag. If
250               provided, it would contain a `dict` with:
251               * `description`: A short description of the target documentation.
252               [CommonMark syntax](https://commonmark.org/) MAY be used for
253               rich text representation.
254               * `url`: The URL for the target documentation. Value MUST be in
255               the form of a URL.
256
257             Read more in the
258             [FastAPI docs for Metadata and Docs URLs]
259             (https://fastapi.tiangolo.com/tutorial/metadata/#metadata-for-tags).
260
261             **Example**
262
263             ```python
264             from fastapi import FastAPI
265
266             tags_metadata = [
267                 {
268                     "name": "users",
269                     "description": "Operations with users. The **login** logic is also here.",
270                 },
271                 {
272                     "name": "items",
273                     "description": "Manage items. So _fancy_ they have their own docs.",
274                     "externalDocs": {
275                         "description": "Items external docs",
276                         "url": "https://fastapi.tiangolo.com/",
277                     },
278                 },
279             ],
280

```

```

281         ]
282
283         app = FastAPI(openapi_tags=tags_metadata)
284         """
285
286     ),
287     ] = None,
288     servers: Annotated[
289         list[dict[str, str] | Any] | None,
290         Doc(
291             """
292             A `list` of `dict`s with connectivity information to a target server.
293
294             You would use it, for example, if your application is served from
295             different domains and you want to use the same Swagger UI in the
296             browser to interact with each of them (instead of having multiple
297             browser tabs open). Or if you want to leave fixed the possible URLs.
298
299             If the servers `list` is not provided, or is an empty `list`, the
300             `servers` property in the generated OpenAPI will be:
301
302             * a `dict` with a `url` value of the application's mounting point
303             (`root_path`) if it's different from `/.`.
304             * otherwise, the `servers` property will be omitted from the OpenAPI
305             schema.
306
307             Each item in the `list` is a `dict` containing:
308
309             * `url`: A URL to the target host. This URL supports Server Variables
310             and MAY be relative, to indicate that the host location is relative
311             to the location where the OpenAPI document is being served. Variable
312             substitutions will be made when a variable is named in `{`brackets`}`.
313             * `description`: An optional string describing the host designated by
314             the URL. [CommonMark syntax](https://commonmark.org/) MAY be used for
315             rich text representation.
316             * `variables`: A `dict` between a variable name and its value. The value
317             is used for substitution in the server's URL template.
318
319             Read more in the
320             [FastAPI docs for Behind a Proxy](https://fastapi.tiangolo.com/advanced/behind-a-
321             proxy/#additional-servers).
322
323             **Example**
324
325             ```python
326             from fastapi import FastAPI
327
328             app = FastAPI(
329                 servers=[
330                     {"url": "https://stag.example.com", "description": "Staging environment"},
331                     {"url": "https://prod.example.com", "description": "Production environment"},
332                 ]
333             )
334             """
335
336     ),
337     ] = None,
338     dependencies: Annotated[
339         Sequence[Depends] | None,
340         Doc(
341             """
342             A list of global dependencies, they will be applied to each
343             *path operation*, including in sub-routers.
344
345             Read more about it in the
346             [FastAPI docs for Global Dependencies]
347             (https://fastapi.tiangolo.com/tutorial/dependencies/global-dependencies/).
348
349             **Example**
350
351             ```python
352             from fastapi import Depends, FastAPI
353
354             from .dependencies import func_dep_1, func_dep_2
355
356             app = FastAPI(dependencies=[Depends(func_dep_1), Depends(func_dep_2)])

```

```

357     """
358     """
359     ),
360     ] = None,
361     default_response_class: Annotated[
362         type[Response],
363         Doc(
364             """
365             The default response class to be used.
366
367             Read more in the
368             [FastAPI docs for Custom Response - HTML, Stream, File, others]
369             (https://fastapi.tiangolo.com/advanced/custom-response/#default-response-class).
370
371             **Example**
372
373             ```python
374             from fastapi import FastAPI
375             from fastapi.responses import ORJSONResponse
376
377             app = FastAPI(default_response_class=ORJSONResponse)
378             ```
379             """
380         ),
381     ] = Default(JSONResponse),
382     redirect_slashes: Annotated[
383         bool,
384         Doc(
385             """
386             Whether to detect and redirect slashes in URLs when the client doesn't
387             use the same format.
388
389             **Example**
390
391             ```python
392             from fastapi import FastAPI
393
394             app = FastAPI(redirect_slashes=True) # the default
395
396             @app.get("/items/")
397             async def read_items():
398                 return [{"item_id": "Foo"}]
399             ```
400
401             With this app, if a client goes to `/items` (without a trailing slash),
402             they will be automatically redirected with an HTTP status code of 307
403             to `/items/`.
404             """
405         ),
406     ] = True,
407     docs_url: Annotated[
408         str | None,
409         Doc(
410             """
411             The path to the automatic interactive API documentation.
412             It is handled in the browser by Swagger UI.
413
414             The default URL is `/docs`. You can disable it by setting it to `None`.
415
416             If `openapi_url` is set to `None`, this will be automatically disabled.
417
418             Read more in the
419             [FastAPI docs for Metadata and Docs URLs]
420             (https://fastapi.tiangolo.com/tutorial/metadata/#docs-urls).
421
422             **Example**
423
424             ```python
425             from fastapi import FastAPI
426
427             app = FastAPI(docs_url="/documentation", redoc_url=None)
428             ```
429             """
430         ),
431     ] = "/docs",
432     redoc_url: Annotated[

```



```

433         str | None,
434     Doc(
435         """
436         The path to the alternative automatic interactive API documentation
437         provided by ReDoc.
438
439         The default URL is `/redoc`. You can disable it by setting it to `None`.
440
441         If `openapi_url` is set to `None`, this will be automatically disabled.
442
443         Read more in the
444         [FastAPI docs for Metadata and Docs URLs]
445         (https://fastapi.tiangolo.com/tutorial/metadata/#docs-urls).
446
447         **Example**
448
449         ```python
450         from fastapi import FastAPI
451
452         app = FastAPI(docs_url="/documentation", redoc_url="redocumentation")
453         """
454     ),
455     ] = "/redoc",
456     swagger_ui_oauth2_redirect_url: Annotated[
457         str | None,
458     Doc(
459         """
460         The OAuth2 redirect endpoint for the Swagger UI.
461
462         By default it is `/docs/oauth2-redirect`.
463
464         This is only used if you use OAuth2 (with the "Authorize" button)
465         with Swagger UI.
466         """
467     ),
468     ] = "/docs/oauth2-redirect",
469     swagger_ui_init_oauth: Annotated[
470         dict[str, Any] | None,
471     Doc(
472         """
473         OAuth2 configuration for the Swagger UI, by default shown at `/docs`.
474
475         Read more about the available configuration options in the
476         [Swagger UI docs](https://swagger.io/docs/open-source-tools/swagger-
477         ui/usage/oauth2/).
478         """
479     ),
480     ] = None,
481     middleware: Annotated[
482         Sequence[Middleware] | None,
483     Doc(
484         """
485         List of middleware to be added when creating the application.
486
487         In FastAPI you would normally do this with `app.add_middleware()`
488         instead.
489
490         Read more in the
491         [FastAPI docs for Middleware](https://fastapi.tiangolo.com/tutorial/middleware/).
492         """
493     ),
494     ] = None,
495     exception_handlers: Annotated[
496         dict[
497             int | type[Exception],
498             Callable[[Request, Any], Coroutine[Any, Any, Response]],
499         ]
500         | None,
501     Doc(
502         """
503         A dictionary with handlers for exceptions.
504
505         In FastAPI, you would normally use the decorator
506         `@app.exception_handler()`.
507         """
508     ),

```

```

509         Read more in the
510         [FastAPI docs for Handling Errors](https://fastapi.tiangolo.com/tutorial/handling-
511         errors/).
512         """
513     ),
514     ] = None,
515     on_startup: Annotated[
516         Sequence[Callable[[], Any]] | None,
517         Doc(
518             """
519             A list of startup event handler functions.
520
521             You should instead use the `lifespan` handlers.
522
523             Read more in the [FastAPI docs for `lifespan`]
524             (https://fastapi.tiangolo.com/advanced/events/).
525             """
526         ),
527     ] = None,
528     on_shutdown: Annotated[
529         Sequence[Callable[[], Any]] | None,
530         Doc(
531             """
532             A list of shutdown event handler functions.
533
534             You should instead use the `lifespan` handlers.
535
536             Read more in the
537             [FastAPI docs for `lifespan`](https://fastapi.tiangolo.com/advanced/events/).
538             """
539         ),
540     ] = None,
541     lifespan: Annotated[
542         Lifespan[AppType] | None,
543         Doc(
544             """
545             A `Lifespan` context manager handler. This replaces `startup` and
546             `shutdown` functions with a single context manager.
547
548             Read more in the
549             [FastAPI docs for `lifespan`](https://fastapi.tiangolo.com/advanced/events/).
550             """
551         ),
552     ] = None,
553     terms_of_service: Annotated[
554         str | None,
555         Doc(
556             """
557             A URL to the Terms of Service for your API.
558
559             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
560
561             Read more at the
562             [FastAPI docs for Metadata and Docs URLs]
563             (https://fastapi.tiangolo.com/tutorial/metadata/#metadata-for-api).
564
565             **Example**
566
567             ```python
568             app = FastAPI(terms_of_service="http://example.com/terms/")
569             ```
570             """
571         ),
572     ] = None,
573     contact: Annotated[
574         dict[str, str] | Any | None,
575         Doc(
576             """
577             A dictionary with the contact information for the exposed API.
578
579             It can contain several fields.
580
581             * `name`: (`str`) The name of the contact person/organization.
582             * `url`: (`str`) A URL pointing to the contact information. MUST be in
583               the format of a URL.
584             * `email`: (`str`) The email address of the contact person/organization.

```

```

585         MUST be in the format of an email address.
586
587         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
588
589         Read more at the
590         [FastAPI docs for Metadata and Docs URLs]
591         (https://fastapi.tiangolo.com/tutorial/metadata/#metadata-for-api).
592
593         **Example**
594
595         ```python
596         app = FastAPI(
597             contact={
598                 "name": "Deadpoolio the Amazing",
599                 "url": "http://x-force.example.com/contact/",
600                 "email": "dp@x-force.example.com",
601             }
602         )
603         """
604
605     ),
606     ] = None,
607     license_info: Annotated[
608         dict[str, str | Any] | None,
609         Doc(
610             """
611             A dictionary with the license information for the exposed API.
612
613             It can contain several fields.
614
615             * `name`: (`str`) **REQUIRED** (if a `license_info` is set). The
616               license name used for the API.
617             * `identifier`: (`str`) An [SPDX](https://spdx.dev/) license expression
618               for the API. The `identifier` field is mutually exclusive of the `url`
619               field. Available since OpenAPI 3.1.0, FastAPI 0.99.0.
620             * `url`: (`str`) A URL to the license used for the API. This MUST be
621               the format of a URL.
622
623             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
624
625             Read more at the
626             [FastAPI docs for Metadata and Docs URLs]
627             (https://fastapi.tiangolo.com/tutorial/metadata/#metadata-for-api).
628
629             **Example**
630
631             ```python
632             app = FastAPI(
633                 license_info={
634                     "name": "Apache 2.0",
635                     "url": "https://www.apache.org/licenses/LICENSE-2.0.html",
636                 }
637             )
638             """
639         ),
640     ] = None,
641     openapi_prefix: Annotated[
642         str,
643         Doc(
644             """
645             A URL prefix for the OpenAPI URL.
646             """
647         ),
648     ],
649     deprecated(
650         """
651         "openapi_prefix" has been deprecated in favor of "root_path", which
652         follows more closely the ASGI standard, is simpler, and more
653         automatic.
654         """
655     ),
656     ] = "",
657     root_path: Annotated[
658         str,
659         Doc(
660             """

```

```

661         A path prefix handled by a proxy that is not seen by the application
662         but is seen by external clients, which affects things like Swagger UI.
663
664         Read more about it at the
665         [FastAPI docs for Behind a Proxy](https://fastapi.tiangolo.com/advanced/behind-a-
666         proxy/).
667
668         **Example**
669
670         ```python
671         from fastapi import FastAPI
672
673         app = FastAPI(root_path="/api/v1")
674         """
675
676     ),
677     ] = "",
678     root_path_in_servers: Annotated[
679         bool,
680         Doc(
681             """
682             To disable automatically generating the URLs in the `servers` field
683             in the autogenerated OpenAPI using the `root_path`.
684
685             Read more about it in the
686             [FastAPI docs for Behind a Proxy](https://fastapi.tiangolo.com/advanced/behind-a-
687             proxy/#disable-automatic-server-from-root-path).
688
689             **Example**
690
691             ```python
692             from fastapi import FastAPI
693
694             app = FastAPI(root_path_in_servers=False)
695             """
696         ),
697     ] = True,
698     responses: Annotated[
699         dict[int | str, dict[str, Any]] | None,
700         Doc(
701             """
702             Additional responses to be shown in OpenAPI.
703
704             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
705
706             Read more about it in the
707             [FastAPI docs for Additional Responses in OpenAPI]
708             (https://fastapi.tiangolo.com/advanced/additional-responses/).
709
710             And in the
711             [FastAPI docs for Bigger Applications](https://fastapi.tiangolo.com/tutorial/bigger-
712             applications/#include-an-apirouter-with-a-custom-prefix-tags-responses-and-dependencies).
713             """
714         ),
715     ] = None,
716     callbacks: Annotated[
717         list[BaseRoute] | None,
718         Doc(
719             """
720             OpenAPI callbacks that should apply to all *path operations*.
721
722             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
723
724             Read more about it in the
725             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
726             callbacks/).
727             """
728         ),
729     ] = None,
730     webhooks: Annotated[
731         routing.APIRouter | None,
732         Doc(
733             """
734             Add OpenAPI webhooks. This is similar to `callbacks` but it doesn't
735             depend on specific *path operations*.
736

```

```

737         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
738
739         **Note**: This is available since OpenAPI 3.1.0, FastAPI 0.99.0.
740
741         Read more about it in the
742         [FastAPI docs for OpenAPI Webhooks](https://fastapi.tiangolo.com/advanced/openapi-
743         webhooks/).
744         """
745     ),
746     ] = None,
747     deprecated: Annotated[
748         bool | None,
749         Doc(
750             """
751             Mark all *path operations* as deprecated. You probably don't need it,
752             but it's available.
753
754             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
755
756             Read more about it in the
757             [FastAPI docs for Path Operation Configuration]
758             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#deprecate-a-path-operation).
759             """
760         ),
761     ] = None,
762     include_in_schema: Annotated[
763         bool,
764         Doc(
765             """
766             To include (or not) all the *path operations* in the generated OpenAPI.
767             You probably don't need it, but it's available.
768
769             This affects the generated OpenAPI (e.g. visible at `/docs`).
770
771             Read more about it in the
772             [FastAPI docs for Query Parameters and String Validations]
773             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
774             openapi).
775             """
776         ),
777     ] = True,
778     swagger_ui_parameters: Annotated[
779         dict[str, Any] | None,
780         Doc(
781             """
782             Parameters to configure Swagger UI, the autogenerated interactive API
783             documentation (by default at `/docs`).
784
785             Read more about it in the
786             [FastAPI docs about how to Configure Swagger UI](https://fastapi.tiangolo.com/how-
787             to/configure-swagger-ui/).
788             """
789         ),
790     ] = None,
791     generate_unique_id_function: Annotated[
792         Callable[[routing.APIRoute], str],
793         Doc(
794             """
795             Customize the function used to generate unique IDs for the *path
796             operations* shown in the generated OpenAPI.
797
798             This is particularly useful when automatically generating clients or
799             SDKs for your API.
800
801             Read more about it in the
802             [FastAPI docs about how to Generate Clients]
803             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
804             """
805         ),
806     ] = Default(generate_unique_id),
807     separate_input_output_schemas: Annotated[
808         bool,
809         Doc(
810             """
811             Whether to generate separate OpenAPI schemas for request body and
812

```

```

813         response body when the results would be more precise.
814
815         This is particularly useful when automatically generating clients.
816
817         For example, if you have a model like:
818
819         ```python
820         from pydantic import BaseModel
821
822         class Item(BaseModel):
823             name: str
824             tags: list[str] = []
825         ```
826
827         When `Item` is used for input, a request body, `tags` is not required,
828         the client doesn't have to provide it.
829
830         But when using `Item` for output, for a response body, `tags` is always
831         available because it has a default value, even if it's just an empty
832         list. So, the client should be able to always expect it.
833
834         In this case, there would be two different schemas, one for input and
835         another one for output.
836
837         Read more about it in the
838         [FastAPI docs about how to separate schemas for input and output]
839         (https://fastapi.tiangolo.com/how-to/separate-openapi-schemas)
840         """
841     ),
842     ] = True,
843     openapi_external_docs: Annotated[
844         dict[str, Any] | None,
845         Doc(
846             """
847             This field allows you to provide additional external documentation links.
848             If provided, it must be a dictionary containing:
849
850             * `description`: A brief description of the external documentation.
851             * `url`: The URL pointing to the external documentation. The value MUST
852             be a valid URL format.
853
854             Example:
855
856             ```python
857             from fastapi import FastAPI
858
859             external_docs = {
860                 "description": "Detailed API Reference",
861                 "url": "https://example.com/api-docs",
862             }
863
864             app = FastAPI(openapi_external_docs=external_docs)
865             """
866         ),
867     ] = None,
868     strict_content_type: Annotated[
869         bool,
870         Doc(
871             """
872             Enable strict checking for request Content-Type headers.
873
874             When `True` (the default), requests with a body that do not include
875             a `Content-Type` header will not be parsed as JSON.
876
877             This prevents potential cross-site request forgery (CSRF) attacks
878             that exploit the browser's ability to send requests without a
879             Content-Type header, bypassing CORS preflight checks. In particular
880             applicable for apps that need to be run locally (in localhost).
881
882             When `False`, requests without a `Content-Type` header will have
883             their body parsed as JSON, which maintains compatibility with
884             certain clients that don't send `Content-Type` headers.
885
886             Read more about it in the
887
888

```

```

889         [FastAPI docs for Strict Content-Type](https://fastapi.tiangolo.com/advanced/strict-
890         content-type/).
891         """
892     ),
893     ] = True,
894     **extra: Annotated[
895         Any,
896         Doc(
897             """
898             Extra keyword arguments to be stored in the app, not used by FastAPI
899             anywhere.
900             """
901         ),
902     ],
903 ) -> None:
904     self.debug = debug
905     self.title = title
906     self.summary = summary
907     self.description = description
908     self.version = version
909     self.terms_of_service = terms_of_service
910     self.contact = contact
911     self.license_info = license_info
912     self.openapi_url = openapi_url
913     self.openapi_tags = openapi_tags
914     self.root_path_in_servers = root_path_in_servers
915     self.docs_url = docs_url
916     self.redoc_url = redoc_url
917     self.swagger_ui_oauth2_redirect_url = swagger_ui_oauth2_redirect_url
918     self.swagger_ui_init_oauth = swagger_ui_init_oauth
919     self.swagger_ui_parameters = swagger_ui_parameters
920     self.servers = servers or []
921     self.separate_input_output_schemas = separate_input_output_schemas
922     self.openapi_external_docs = openapi_external_docs
923     self.extra = extra
924     self.openapi_version: Annotated[
925         str,
926         Doc(
927             """
928             The version string of OpenAPI.
929
930             FastAPI will generate OpenAPI version 3.1.0, and will output that as
931             the OpenAPI version. But some tools, even though they might be
932             compatible with OpenAPI 3.1.0, might not recognize it as a valid.
933
934             So you could override this value to trick those tools into using
935             the generated OpenAPI. Have in mind that this is a hack. But if you
936             avoid using features added in OpenAPI 3.1.0, it might work for your
937             use case.
938
939             This is not passed as a parameter to the `FastAPI` class to avoid
940             giving the false idea that FastAPI would generate a different OpenAPI
941             schema. It is only available as an attribute.
942
943             **Example**
944
945             ```python
946             from fastapi import FastAPI
947
948             app = FastAPI()
949
950             app.openapi_version = "3.0.2"
951             ```
952             """
953         ),
954     ] = "3.1.0"
955     self.openapi_schema: dict[str, Any] | None = None
956     if self.openapi_url:
957         assert self.title, "A title must be provided for OpenAPI, e.g.: 'My API'"
958         assert self.version, "A version must be provided for OpenAPI, e.g.: '2.1.0'"
959     # TODO: remove when discarding the openapi_prefix parameter
960     if openapi_prefix:
961         logger.warning(
962             '"openapi_prefix" has been deprecated in favor of "root_path", which '
963             'follows more closely the ASGI standard, is simpler, and more '
964             'automatic. Check the docs at "

```

```

965         "https://fastapi.tiangolo.com/advanced/sub-applications/"
966     )
967     self.webhooks: Annotated[
968         routing.APIRouter,
969         Doc(
970             """
971             The `app.webhooks` attribute is an `APIRouter` with the *path
972             operations* that will be used just for documentation of webhooks.
973
974             Read more about it in the
975             [FastAPI docs for OpenAPI Webhooks](https://fastapi.tiangolo.com/advanced/openapi-
976             webhooks/).
977             """
978         ),
979     ] = webhooks or routing.APIRouter()
980     self.root_path = root_path or openapi_prefix
981     self.state: Annotated[
982         State,
983         Doc(
984             """
985             A state object for the application. This is the same object for the
986             entire application, it doesn't change from request to request.
987
988             You normally wouldn't use this in FastAPI, for most of the cases you
989             would instead use FastAPI dependencies.
990
991             This is simply inherited from Starlette.
992
993             Read more about it in the
994             [Starlette docs for Applications](https://www.starlette.dev/applications/#storing-
995             state-on-the-app-instance).
996             """
997         ),
998     ] = State()
999     self.dependency_overrides: Annotated[
1000         dict[Callable[..., Any], Callable[..., Any]],
1001         Doc(
1002             """
1003             A dictionary with overrides for the dependencies.
1004
1005             Each key is the original dependency callable, and the value is the
1006             actual dependency that should be called.
1007
1008             This is for testing, to replace expensive dependencies with testing
1009             versions.
1010
1011             Read more about it in the
1012             [FastAPI docs for Testing Dependencies with Overrides]
1013             (https://fastapi.tiangolo.com/advanced/testing-dependencies/).
1014             """
1015         ),
1016     ] = {}
1017     self.router: routing.APIRouter = routing.APIRouter(
1018         routes=routes,
1019         redirect_slashes=redirect_slashes,
1020         dependency_overrides_provider=self,
1021         on_startup=on_startup,
1022         on_shutdown=on_shutdown,
1023         lifespan=lifespan,
1024         default_response_class=default_response_class,
1025         dependencies=dependencies,
1026         callbacks=callbacks,
1027         deprecated=deprecated,
1028         include_in_schema=include_in_schema,
1029         responses=responses,
1030         generate_unique_id_function=generate_unique_id_function,
1031         strict_content_type=strict_content_type,
1032     )
1033     self.exception_handlers: dict[
1034         Any, Callable[[Request, Any], Response | Awaitable[Response]]
1035     ] = {} if exception_handlers is None else dict(exception_handlers)
1036     self.exception_handlers.setdefault(HTTPException, http_exception_handler)
1037     self.exception_handlers.setdefault(
1038         RequestValidationError, request_validation_exception_handler
1039     )

```



```
# Starlette still has incorrect type specification for the handlers
self.exception_handlers.setdefault(
    WebSocketRequestValidationError,
    websocket_request_validation_exception_handler, # type: ignore[arg-type] # ty:
    ignore[unused-ignore-comment]
) # ty: ignore[no-matching-overload]

self.user_middleware: list[Middleware] = (
    [] if middleware is None else list(middleware)
)
self.middleware_stack: ASGIApp | None = None
self.setup()
```

attr `openapi_version` instance-attribute

```
openapi_version = '3.1.0'
```

The version string of OpenAPI.

FastAPI will generate OpenAPI version `3.1.0`, and will output that as the OpenAPI version. But some tools, even though they might be compatible with OpenAPI `3.1.0`, might not recognize it as a valid.

So you could override this value to trick those tools into using the generated OpenAPI. Have in mind that this is a hack. But if you avoid using features added in OpenAPI `3.1.0`, it might work for your use case.

This is not passed as a parameter to the `FastAPI` class to avoid giving the false idea that FastAPI would generate a different OpenAPI schema. It is only available as an attribute.

Example

```
from fastapi import FastAPI

app = FastAPI()

app.openapi_version = "3.0.2"
```

attr `webhooks` instance-attribute

```
webhooks = webhooks or APIRouter()
```

The `app.webhooks` attribute is an `APIRouter` with the *path operations* that will be used just for documentation of webhooks.

Read more about it in the [FastAPI docs for OpenAPI Webhooks](#).

attr `state` instance-attribute

```
state = State()
```

A state object for the application. This is the same object for the entire application, it doesn't change from request to request.

You normally wouldn't use this in FastAPI, for most of the cases you would instead use FastAPI dependencies.

This is simply inherited from Starlette.

Read more about it in the [Starlette docs for Applications](#).

attr `dependency_overrides` instance-attribute

```
dependency_overrides = {}
```

A dictionary with overrides for the dependencies.

Each key is the original dependency callable, and the value is the actual dependency that should be called.

This is for testing, to replace expensive dependencies with testing versions.

Read more about it in the [FastAPI docs for Testing Dependencies with Overrides](#).

meth `openapi`

```
openapi()
```

Generate the OpenAPI schema of the application. This is called by FastAPI internally.

The first time it is called it stores the result in the attribute `app.openapi_schema`, and next times it is called, it just returns that same result. To avoid the cost of generating the schema every time.

If you need to modify the generated OpenAPI schema, you could modify it.

Read more in the [FastAPI docs for OpenAPI](#).

< > Source code in `fastapi/applications.py`

```

1068 def openapi(self) -> dict[str, Any]:
1069     """
1070     Generate the OpenAPI schema of the application. This is called by FastAPI
1071     internally.
1072
1073     The first time it is called it stores the result in the attribute
1074     `app.openapi_schema`, and next times it is called, it just returns that same
1075     result. To avoid the cost of generating the schema every time.
1076
1077     If you need to modify the generated OpenAPI schema, you could modify it.
1078
1079     Read more in the
1080     [FastAPI docs for OpenAPI](https://fastapi.tiangolo.com/how-to/extending-openapi/).
1081     """
1082     if not self.openapi_schema:
1083         self.openapi_schema = get_openapi(
1084             title=self.title,
1085             version=self.version,
1086             openapi_version=self.openapi_version,
1087             summary=self.summary,
1088             description=self.description,
1089             terms_of_service=self.terms_of_service,
1090             contact=self.contact,
1091             license_info=self.license_info,
1092             routes=self.routes,
1093             webhooks=self.webhooks.routes,
1094             tags=self.openapi_tags,
1095             servers=self.servers,
1096             separate_input_output_schemas=self.separate_input_output_schemas,
1097             external_docs=self.openapi_external_docs,
1098         )
1099     return self.openapi_schema

```

meth `websocket`

```
websocket(path, name=None, *, dependencies=None)
```

Decorate a WebSocket function.

Read more about it in the [FastAPI docs for WebSockets](#).

Example

```
from fastapi import FastAPI, WebSocket

app = FastAPI()

@app.websocket("/ws")
async def websocket_endpoint(websocket: WebSocket):
    await websocket.accept()
    while True:
        data = await websocket.receive_text()
        await websocket.send_text(f"Message text was: {data}")
```

PARAMETER	DESCRIPTION
path	WebSocket path. TYPE: str
name	A name for the WebSocket. Only used internally. TYPE: str None DEFAULT: None
dependencies	A list of dependencies (using Depends()) to be used for this WebSocket. Read more about it in the FastAPI docs for WebSockets . TYPE: Sequence[Depends] None DEFAULT: None

<> Source code in `fastapi/applications.py`

```

1293 def websocket(
1294     self,
1295     path: Annotated[
1296         str,
1297         Doc(
1298             """
1299             WebSocket path.
1300             """
1301         ),
1302     ],
1303     name: Annotated[
1304         str | None,
1305         Doc(
1306             """
1307             A name for the WebSocket. Only used internally.
1308             """
1309         ),
1310     ] = None,
1311     *,
1312     dependencies: Annotated[
1313         Sequence[Depends] | None,
1314         Doc(
1315             """
1316             A list of dependencies (using `Depends()`) to be used for this
1317             WebSocket.
1318
1319             Read more about it in the
1320             [FastAPI docs for WebSockets](https://fastapi.tiangolo.com/advanced/websockets/).
1321             """
1322         ),
1323     ] = None,
1324 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
1325     """
1326     Decorate a WebSocket function.
1327
1328     Read more about it in the
1329     [FastAPI docs for WebSockets](https://fastapi.tiangolo.com/advanced/websockets/).
1330
1331     **Example**
1332
1333     ```python
1334     from fastapi import FastAPI, WebSocket
1335
1336     app = FastAPI()
1337
1338     @app.websocket("/ws")
1339     async def websocket_endpoint(websocket: WebSocket):
1340         await websocket.accept()
1341         while True:
1342             data = await websocket.receive_text()
1343             await websocket.send_text(f"Message text was: {data}")
1344     ```
1345     """
1346
1347     def decorator(func: DecoratedCallable) -> DecoratedCallable:
1348         self.add_api_websocket_route(
1349             path,
1350             func,
1351             name=name,
1352             dependencies=dependencies,
1353         )
1354         return func
1355
1356     return decorator

```

meth `include_router`

```

include_router(
    router,

```

```
*,
prefix="",
tags=None,
dependencies=None,
responses=None,
deprecated=None,
include_in_schema=True,
default_response_class=Default(JSONResponse),
callbacks=None,
generate_unique_id_function=Default(generate_unique_id)
)
```

Include an `APIRouter` in the same app.

Read more about it in the [FastAPI docs for Bigger Applications](#).

EXAMPLE

```
from fastapi import FastAPI

from .users import users_router

app = FastAPI()

app.include_router(users_router)
```

PARAMETER	DESCRIPTION
<code>router</code>	The <code>APIRouter</code> to include. TYPE: <code>APIRouter</code>
<code>prefix</code>	An optional path prefix for the router. TYPE: <code>str</code> DEFAULT: <code>''</code>
<code>tags</code>	A list of tags to be applied to all the <i>path operations</i> in this router. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum] None</code> DEFAULT: <code>None</code>
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to all the <i>path operations</i> in this router. Read more about it in the FastAPI docs for Bigger Applications - Multiple Files. Example <pre>from fastapi import Depends, FastAPI from .dependencies import get_token_header from .internal import admin app = FastAPI() app.include_router(admin.router, dependencies=[Depends(get_token_header)],)</pre> TYPE: <code>Sequence[Depends] None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>responses</code>	Additional responses to be shown in OpenAPI. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Additional Responses in OpenAPI. And in the FastAPI docs for Bigger Applications. TYPE: <code>dict[int str, dict[str, Any]] None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark all the <i>path operations</i> in this router as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Example <pre>from fastapi import FastAPI from .internal import old_api app = FastAPI() app.include_router(old_api.router, deprecated=True,)</pre> TYPE: <code>bool None</code> DEFAULT: <code>None</code>
<code>include_in_schema</code>	Include (or not) all the <i>path operations</i> in this router in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Example <pre>from fastapi import FastAPI from .internal import old_api app = FastAPI() app.include_router(old_api.router, include_in_schema=False,)</pre> TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>default_response_class</code>	Default response class to be used for the <i>path operations</i> in this router. Read more in the FastAPI docs for Custom Response - HTML, Stream, File, others. Example <pre>from fastapi import FastAPI from fastapi.responses import ORJSONResponse from .internal import old_api app = FastAPI() app.include_router(</pre>

PARAMETER	DESCRIPTION
	<pre>old_api.router, default_response_class=ORJSONResponse,)</pre> TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/applications.py

```

1358 def include_router(
1359     self,
1360     router: Annotated[routing.APIRouter, Doc("The `APIRouter` to include.")],
1361     *,
1362     prefix: Annotated[str, Doc("An optional path prefix for the router.")] = "",
1363     tags: Annotated[
1364         list[str | Enum] | None,
1365         Doc(
1366             """
1367             A list of tags to be applied to all the *path operations* in this
1368             router.
1369
1370             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1371
1372             Read more about it in the
1373             [FastAPI docs for Path Operation Configuration]
1374             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
1375             """
1376         ),
1377     ] = None,
1378     dependencies: Annotated[
1379         Sequence[Depends] | None,
1380         Doc(
1381             """
1382             A list of dependencies (using `Depends()`) to be applied to all the
1383             *path operations* in this router.
1384
1385             Read more about it in the
1386             [FastAPI docs for Bigger Applications - Multiple Files]
1387             (https://fastapi.tiangolo.com/tutorial/bigger-applications/#include-an-apirouter-with-a-custom-
1388             prefix-tags-responses-and-dependencies).
1389
1390             **Example**
1391
1392             ```python
1393             from fastapi import Depends, FastAPI
1394
1395             from .dependencies import get_token_header
1396             from .internal import admin
1397
1398             app = FastAPI()
1399
1400             app.include_router(
1401                 admin.router,
1402                 dependencies=[Depends(get_token_header)],
1403             )
1404             ...
1405             """
1406         ),
1407     ] = None,
1408     responses: Annotated[
1409         dict[int | str, dict[str, Any]] | None,
1410         Doc(
1411             """
1412             Additional responses to be shown in OpenAPI.
1413
1414             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1415
1416             Read more about it in the
1417             [FastAPI docs for Additional Responses in OpenAPI]
1418             (https://fastapi.tiangolo.com/advanced/additional-responses/).
1419
1420             And in the
1421             [FastAPI docs for Bigger Applications](https://fastapi.tiangolo.com/tutorial/bigger-
1422             applications/#include-an-apirouter-with-a-custom-prefix-tags-responses-and-dependencies).
1423             """
1424         ),
1425     ] = None,
1426     deprecated: Annotated[
1427         bool | None,
1428         Doc(
1429             """

```



```

1430         Mark all the *path operations* in this router as deprecated.
1431
1432         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1433
1434         **Example**
1435
1436         ```python
1437         from fastapi import FastAPI
1438
1439         from .internal import old_api
1440
1441         app = FastAPI()
1442
1443         app.include_router(
1444             old_api.router,
1445             deprecated=True,
1446         )
1447         """
1448
1449     ),
1450     ] = None,
1451     include_in_schema: Annotated[
1452         bool,
1453         Doc(
1454             """
1455             Include (or not) all the *path operations* in this router in the
1456             generated OpenAPI schema.
1457
1458             This affects the generated OpenAPI (e.g. visible at `/docs`).
1459
1460             **Example**
1461
1462             ```python
1463             from fastapi import FastAPI
1464
1465             from .internal import old_api
1466
1467             app = FastAPI()
1468
1469             app.include_router(
1470                 old_api.router,
1471                 include_in_schema=False,
1472             )
1473             """
1474         ),
1475     ] = True,
1476     default_response_class: Annotated[
1477         type[Response],
1478         Doc(
1479             """
1480             Default response class to be used for the *path operations* in this
1481             router.
1482
1483             Read more in the
1484             [FastAPI docs for Custom Response - HTML, Stream, File, others]
1485             (https://fastapi.tiangolo.com/advanced/custom-response/#default-response-class).
1486
1487             **Example**
1488
1489             ```python
1490             from fastapi import FastAPI
1491             from fastapi.responses import ORJSONResponse
1492
1493             from .internal import old_api
1494
1495             app = FastAPI()
1496
1497             app.include_router(
1498                 old_api.router,
1499                 default_response_class=ORJSONResponse,
1500             )
1501             """
1502         ),
1503     ] = Default(JSONResponse),
1504
1505     ] = Default(JSONResponse),

```

```

1506     callbacks: Annotated[
1507         list[BaseRoute] | None,
1508         Doc(
1509             """
1510             List of *path operations* that will be used as OpenAPI callbacks.
1511
1512             This is only for OpenAPI documentation, the callbacks won't be used
1513             directly.
1514
1515             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1516
1517             Read more about it in the
1518             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
1519 callbacks/).
1520             """
1521         ),
1522     ] = None,
1523     generate_unique_id_function: Annotated[
1524         Callable[[routing.APIRoute], str],
1525         Doc(
1526             """
1527             Customize the function used to generate unique IDs for the *path
1528             operations* shown in the generated OpenAPI.
1529
1530             This is particularly useful when automatically generating clients or
1531             SDKs for your API.
1532
1533             Read more about it in the
1534             [FastAPI docs about how to Generate Clients]
1535             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
1536             """
1537         ),
1538     ] = Default(generate_unique_id),
1539 ) -> None:
1540     """
1541     Include an `APIRouter` in the same app.
1542
1543     Read more about it in the
1544     [FastAPI docs for Bigger Applications](https://fastapi.tiangolo.com/tutorial/bigger-
1545 applications/).
1546
1547     ## Example
1548
1549     ```python
1550     from fastapi import FastAPI
1551
1552     from .users import users_router
1553
1554     app = FastAPI()
1555
1556     app.include_router(users_router)
1557     ```
1558     """
1559     self.router.include_router(
1560         router,
1561         prefix=prefix,
1562         tags=tags,
1563         dependencies=dependencies,
1564         responses=responses,
1565         deprecated=deprecated,
1566         include_in_schema=include_in_schema,
1567         default_response_class=default_response_class,
1568         callbacks=callbacks,
1569         generate_unique_id_function=generate_unique_id_function,
1570     )

```

meth get

```

get(
    path,
    *,
    response_model=Default(None),
    status_code=None,

```

```
tags=None,
dependencies=None,
summary=None,
description=None,
response_description="Successful Response",
responses=None,
deprecated=None,
operation_id=None,
response_model_include=None,
response_model_exclude=None,
response_model_by_alias=True,
response_model_exclude_unset=False,
response_model_exclude_defaults=False,
response_model_exclude_none=False,
include_in_schema=True,
response_class=Default(JSONResponse),
name=None,
callbacks=None,
openapi_extra=None,
generate_unique_id_function=Default(generate_unique_id)
)
```

Add a *path operation* using an HTTP GET operation.

EXAMPLE

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/items/")
def read_items():
    return [{"name": "Empanada"}, {"name": "Arepa"}]
```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error

PARAMETER	DESCRIPTION
	from the API developer. So, FastAPI will raise an error and return a 500 error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: Any DEFAULT: Default(None)
status_code	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: int None DEFAULT: None
tags	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: list[str Enum] None DEFAULT: None
dependencies	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: Sequence[Depends] None DEFAULT: None
summary	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: str None DEFAULT: None
description	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: str None DEFAULT: None
response_description	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: str DEFAULT: 'Successful Response'
responses	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>).

PARAMETER	DESCRIPTION		
		TYPE:	DEFAULT:
		<code>dict[int str, dict[str, Any]] None</code>	<code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool None</code> DEFAULT: <code>None</code>		
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str None</code> DEFAULT: <code>None</code>		
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx None</code> DEFAULT: <code>None</code>		
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx None</code> DEFAULT: <code>None</code>		
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>		
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>		
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set		

PARAMETER	DESCRIPTION
	but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/applications.py

```

1563 def get(
1564     self,
1565     path: Annotated[
1566         str,
1567         Doc(
1568             """
1569             The URL path to be used for this *path operation*.
1570
1571             For example, in `http://example.com/items`, the path is `/items`.
1572             """
1573         ),
1574     ],
1575     *,
1576     response_model: Annotated[
1577         Any,
1578         Doc(
1579             """
1580             The type to use for the response.
1581
1582             It could be any valid Pydantic *field* type. So, it doesn't have to
1583             be a Pydantic model, it could be other things, like a `list`, `dict`,
1584             etc.
1585
1586             It will be used for:
1587
1588             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
1589               show it as the response (JSON Schema).
1590             * Serialization: you could return an arbitrary object and the
1591               `response_model` would be used to serialize that object into the
1592               corresponding JSON.
1593             * Filtering: the JSON sent to the client will only contain the data
1594               (fields) defined in the `response_model`. If you returned an object
1595               that contains an attribute `password` but the `response_model` does
1596               not include that field, the JSON sent to the client would not have
1597               that `password`.
1598             * Validation: whatever you return will be serialized with the
1599               `response_model`, converting any data as necessary to generate the
1600               corresponding JSON. But if the data in the object returned is not
1601               valid, that would mean a violation of the contract with the client,
1602               so it's an error from the API developer. So, FastAPI will raise an
1603               error and return a 500 error code (Internal Server Error).
1604
1605               Read more about it in the
1606               [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
1607 model/).
1608             """
1609         ),
1610     ] = Default(None),
1611     status_code: Annotated[
1612         int | None,
1613         Doc(
1614             """
1615             The default status code to be used for the response.
1616
1617             You could override the status code by returning a response directly.
1618
1619             Read more about it in the
1620             [FastAPI docs for Response Status Code]
1621             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
1622             """
1623         ),
1624     ] = None,
1625     tags: Annotated[
1626         list[str | Enum] | None,
1627         Doc(
1628             """
1629             A list of tags to be applied to the *path operation*.
1630
1631             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1632
1633             Read more about it in the
1634

```



```

1635         [FastAPI docs for Path Operation Configuration]
1636         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
1637         """
1638     ),
1639     ] = None,
1640     dependencies: Annotated[
1641         Sequence[Depends] | None,
1642         Doc(
1643             """
1644             A list of dependencies (using `Depends()`) to be applied to the
1645             *path operation*.
1646
1647             Read more about it in the
1648             [FastAPI docs for Dependencies in path operation decorators]
1649             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
1650             """
1651         ),
1652     ] = None,
1653     summary: Annotated[
1654         str | None,
1655         Doc(
1656             """
1657             A summary for the *path operation*.
1658
1659             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1660
1661             Read more about it in the
1662             [FastAPI docs for Path Operation Configuration]
1663             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
1664             """
1665         ),
1666     ] = None,
1667     description: Annotated[
1668         str | None,
1669         Doc(
1670             """
1671             A description for the *path operation*.
1672
1673             If not provided, it will be extracted automatically from the docstring
1674             of the *path operation function*.
1675
1676             It can contain Markdown.
1677
1678             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1679
1680             Read more about it in the
1681             [FastAPI docs for Path Operation Configuration]
1682             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
1683             """
1684         ),
1685     ] = None,
1686     response_description: Annotated[
1687         str,
1688         Doc(
1689             """
1690             The description for the default response.
1691
1692             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1693             """
1694         ),
1695     ] = "Successful Response",
1696     responses: Annotated[
1697         dict[int | str, dict[str, Any]] | None,
1698         Doc(
1699             """
1700             Additional responses that could be returned by this *path operation*.
1701
1702             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1703             """
1704         ),
1705     ] = None,
1706     deprecated: Annotated[
1707         bool | None,
1708         Doc(
1709             """
1710             Mark this *path operation* as deprecated.

```

```

1711         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1712         """
1713     ),
1714 ] = None,
1715 operation_id: Annotated[
1716     str | None,
1717     Doc(
1718         """
1719         Custom operation ID to be used by this *path operation*.
1720
1721         By default, it is generated automatically.
1722
1723         If you provide a custom operation ID, you need to make sure it is
1724         unique for the whole API.
1725
1726         You can customize the
1727         operation ID generation with the parameter
1728         `generate_unique_id_function` in the `FastAPI` class.
1729
1730         Read more about it in the
1731         [FastAPI docs about how to Generate Clients]
1732         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
1733         """
1734     ),
1735 ] = None,
1736 response_model_include: Annotated[
1737     IncEx | None,
1738     Doc(
1739         """
1740         Configuration passed to Pydantic to include only certain fields in the
1741         response data.
1742
1743         Read more about it in the
1744         [FastAPI docs for Response Model - Return Type]
1745         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
1746         """
1747     ),
1748 ] = None,
1749 response_model_exclude: Annotated[
1750     IncEx | None,
1751     Doc(
1752         """
1753         Configuration passed to Pydantic to exclude certain fields in the
1754         response data.
1755
1756         Read more about it in the
1757         [FastAPI docs for Response Model - Return Type]
1758         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
1759         """
1760     ),
1761 ] = None,
1762 response_model_by_alias: Annotated[
1763     bool,
1764     Doc(
1765         """
1766         Configuration passed to Pydantic to define if the response model
1767         should be serialized by alias when an alias is used.
1768
1769         Read more about it in the
1770         [FastAPI docs for Response Model - Return Type]
1771         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
1772         """
1773     ),
1774 ] = True,
1775 response_model_exclude_unset: Annotated[
1776     bool,
1777     Doc(
1778         """
1779         Configuration passed to Pydantic to define if the response data
1780         should have all the fields, including the ones that were not set and
1781         have their default values. This is different from
1782         `response_model_exclude_defaults` in that if the fields are set,

```

```

1787         they will be included in the response, even if the value is the same
1788         as the default.
1789
1790         When `True`, default values are omitted from the response.
1791
1792         Read more about it in the
1793         [FastAPI docs for Response Model - Return Type]
1794         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
1795         parameter).
1796         """
1797     ),
1798     ] = False,
1799     response_model_exclude_defaults: Annotated[
1800         bool,
1801         Doc(
1802             """
1803             Configuration passed to Pydantic to define if the response data
1804             should have all the fields, including the ones that have the same value
1805             as the default. This is different from `response_model_exclude_unset`
1806             in that if the fields are set but contain the same default values,
1807             they will be excluded from the response.
1808
1809             When `True`, default values are omitted from the response.
1810
1811             Read more about it in the
1812             [FastAPI docs for Response Model - Return Type]
1813             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
1814             parameter).
1815             """
1816         ),
1817     ] = False,
1818     response_model_exclude_none: Annotated[
1819         bool,
1820         Doc(
1821             """
1822             Configuration passed to Pydantic to define if the response data should
1823             exclude fields set to `None`.
1824
1825             This is much simpler (less smart) than `response_model_exclude_unset`
1826             and `response_model_exclude_defaults`. You probably want to use one of
1827             those two instead of this one, as those allow returning `None` values
1828             when it makes sense.
1829
1830             Read more about it in the
1831             [FastAPI docs for Response Model - Return Type]
1832             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
1833             """
1834         ),
1835     ] = False,
1836     include_in_schema: Annotated[
1837         bool,
1838         Doc(
1839             """
1840             Include this *path operation* in the generated OpenAPI schema.
1841
1842             This affects the generated OpenAPI (e.g. visible at `/docs`).
1843
1844             Read more about it in the
1845             [FastAPI docs for Query Parameters and String Validations]
1846             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
1847             openapi).
1848             """
1849         ),
1850     ] = True,
1851     response_class: Annotated[
1852         type[Response],
1853         Doc(
1854             """
1855             Response class to be used for this *path operation*.
1856
1857             This will not be used if you return a response directly.
1858
1859             Read more about it in the
1860             [FastAPI docs for Custom Response - HTML, Stream, File, others]
1861             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
1862             """

```

```

1863     ),
1864     ] = Default(JSONResponse),
1865     name: Annotated[
1866         str | None,
1867         Doc(
1868             """
1869             Name for this *path operation*. Only used internally.
1870             """
1871         ),
1872     ] = None,
1873     callbacks: Annotated[
1874         list[BaseRoute] | None,
1875         Doc(
1876             """
1877             List of *path operations* that will be used as OpenAPI callbacks.
1878
1879             This is only for OpenAPI documentation, the callbacks won't be used
1880             directly.
1881
1882             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
1883
1884             Read more about it in the
1885             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
1886             callbacks/).
1887             """
1888         ),
1889     ] = None,
1890     openapi_extra: Annotated[
1891         dict[str, Any] | None,
1892         Doc(
1893             """
1894             Extra metadata to be included in the OpenAPI schema for this *path
1895             operation*.
1896
1897             Read more about it in the
1898             [FastAPI docs for Path Operation Advanced Configuration]
1899             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
1900             path-operation-schema).
1901             """
1902         ),
1903     ] = None,
1904     generate_unique_id_function: Annotated[
1905         Callable[[routing.APIRoute], str],
1906         Doc(
1907             """
1908             Customize the function used to generate unique IDs for the *path
1909             operations* shown in the generated OpenAPI.
1910
1911             This is particularly useful when automatically generating clients or
1912             SDKs for your API.
1913
1914             Read more about it in the
1915             [FastAPI docs about how to Generate Clients]
1916             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
1917             """
1918         ),
1919     ] = Default(generate_unique_id),
1920 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
1921     """
1922     Add a *path operation* using an HTTP GET operation.
1923
1924     ## Example
1925
1926     ```python
1927     from fastapi import FastAPI
1928
1929     app = FastAPI()
1930
1931     @app.get("/items/")
1932     def read_items():
1933         return [{"name": "Empanada"}, {"name": "Arepa"}]
1934     """
1935     return self.router.get(
1936         path,
1937         response_model=response_model,

```

```

        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )

```

meth put

```

put(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP PUT operation.

EXAMPLE

```

from fastapi import FastAPI
from pydantic import BaseModel

class Item(BaseModel):
    name: str
    description: str | None = None

app = FastAPI()

@app.put("/items/{item_id}")
def replace_item(item_id: str, item: Item):
    return {"message": "Item replaced", "id": item_id}

```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: Any DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> None DEFAULT: None
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum]</code> None DEFAULT: None
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends]</code> None DEFAULT: None

PARAMETER	DESCRIPTION
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>

PARAMETER	DESCRIPTION
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/applications.py

```

1936 def put(
1937     self,
1938     path: Annotated[
1939         str,
1940         Doc(
1941             """
1942             The URL path to be used for this *path operation*.
1943
1944             For example, in `http://example.com/items`, the path is `/items`.
1945             """
1946         ),
1947     ],
1948     *,
1949     response_model: Annotated[
1950         Any,
1951         Doc(
1952             """
1953             The type to use for the response.
1954
1955             It could be any valid Pydantic *field* type. So, it doesn't have to
1956             be a Pydantic model, it could be other things, like a `list`, `dict`,
1957             etc.
1958
1959             It will be used for:
1960
1961             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
1962               show it as the response (JSON Schema).
1963             * Serialization: you could return an arbitrary object and the
1964               `response_model` would be used to serialize that object into the
1965               corresponding JSON.
1966             * Filtering: the JSON sent to the client will only contain the data
1967               (fields) defined in the `response_model`. If you returned an object
1968               that contains an attribute `password` but the `response_model` does
1969               not include that field, the JSON sent to the client would not have
1970               that `password`.
1971             * Validation: whatever you return will be serialized with the
1972               `response_model`, converting any data as necessary to generate the
1973               corresponding JSON. But if the data in the object returned is not
1974               valid, that would mean a violation of the contract with the client,
1975               so it's an error from the API developer. So, FastAPI will raise an
1976               error and return a 500 error code (Internal Server Error).
1977
1978             Read more about it in the
1979             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
1980             model/).
1981             """
1982         ),
1983     ] = Default(None),
1984     status_code: Annotated[
1985         int | None,
1986         Doc(
1987             """
1988             The default status code to be used for the response.
1989
1990             You could override the status code by returning a response directly.
1991
1992             Read more about it in the
1993             [FastAPI docs for Response Status Code]
1994             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
1995             """
1996         ),
1997     ] = None,
1998     tags: Annotated[
1999         list[str | Enum] | None,
2000         Doc(
2001             """
2002             A list of tags to be applied to the *path operation*.
2003
2004             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2005
2006             Read more about it in the
2007

```

```

2008         [FastAPI docs for Path Operation Configuration]
2009         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
2010         """
2011     ),
2012     ] = None,
2013     dependencies: Annotated[
2014         Sequence[Depends] | None,
2015         Doc(
2016             """
2017             A list of dependencies (using `Depends()`) to be applied to the
2018             *path operation*.
2019
2020             Read more about it in the
2021             [FastAPI docs for Dependencies in path operation decorators]
2022             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
2023             """
2024         ),
2025     ] = None,
2026     summary: Annotated[
2027         str | None,
2028         Doc(
2029             """
2030             A summary for the *path operation*.
2031
2032             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2033
2034             Read more about it in the
2035             [FastAPI docs for Path Operation Configuration]
2036             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
2037             """
2038         ),
2039     ] = None,
2040     description: Annotated[
2041         str | None,
2042         Doc(
2043             """
2044             A description for the *path operation*.
2045
2046             If not provided, it will be extracted automatically from the docstring
2047             of the *path operation function*.
2048
2049             It can contain Markdown.
2050
2051             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2052
2053             Read more about it in the
2054             [FastAPI docs for Path Operation Configuration]
2055             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
2056             """
2057         ),
2058     ] = None,
2059     response_description: Annotated[
2060         str,
2061         Doc(
2062             """
2063             The description for the default response.
2064
2065             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2066             """
2067         ),
2068     ] = "Successful Response",
2069     responses: Annotated[
2070         dict[int | str, dict[str, Any]] | None,
2071         Doc(
2072             """
2073             Additional responses that could be returned by this *path operation*.
2074
2075             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2076             """
2077         ),
2078     ] = None,
2079     deprecated: Annotated[
2080         bool | None,
2081         Doc(
2082             """
2083             Mark this *path operation* as deprecated.

```

```

2084         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2085         """
2086     ),
2087 ] = None,
2088 operation_id: Annotated[
2089     str | None,
2090     Doc(
2091         """
2092         Custom operation ID to be used by this *path operation*.
2093
2094         By default, it is generated automatically.
2095
2096         If you provide a custom operation ID, you need to make sure it is
2097         unique for the whole API.
2098
2099         You can customize the
2100         operation ID generation with the parameter
2101         `generate_unique_id_function` in the `FastAPI` class.
2102
2103         Read more about it in the
2104         [FastAPI docs about how to Generate Clients]
2105         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
2106         """
2107     ),
2108 ] = None,
2109 response_model_include: Annotated[
2110     IncEx | None,
2111     Doc(
2112         """
2113         Configuration passed to Pydantic to include only certain fields in the
2114         response data.
2115
2116         Read more about it in the
2117         [FastAPI docs for Response Model - Return Type]
2118         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2119         """
2120     ),
2121 ] = None,
2122 response_model_exclude: Annotated[
2123     IncEx | None,
2124     Doc(
2125         """
2126         Configuration passed to Pydantic to exclude certain fields in the
2127         response data.
2128
2129         Read more about it in the
2130         [FastAPI docs for Response Model - Return Type]
2131         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2132         """
2133     ),
2134 ] = None,
2135 response_model_by_alias: Annotated[
2136     bool,
2137     Doc(
2138         """
2139         Configuration passed to Pydantic to define if the response model
2140         should be serialized by alias when an alias is used.
2141
2142         Read more about it in the
2143         [FastAPI docs for Response Model - Return Type]
2144         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2145         """
2146     ),
2147 ] = True,
2148 response_model_exclude_unset: Annotated[
2149     bool,
2150     Doc(
2151         """
2152         Configuration passed to Pydantic to define if the response data
2153         should have all the fields, including the ones that were not set and
2154         have their default values. This is different from
2155         `response_model_exclude_defaults` in that if the fields are set,

```

```

2160         they will be included in the response, even if the value is the same
2161         as the default.
2162
2163         When `True`, default values are omitted from the response.
2164
2165         Read more about it in the
2166         [FastAPI docs for Response Model - Return Type]
2167         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2168         parameter).
2169         """
2170     ),
2171     ] = False,
2172     response_model_exclude_defaults: Annotated[
2173         bool,
2174         Doc(
2175             """
2176             Configuration passed to Pydantic to define if the response data
2177             should have all the fields, including the ones that have the same value
2178             as the default. This is different from `response_model_exclude_unset`
2179             in that if the fields are set but contain the same default values,
2180             they will be excluded from the response.
2181
2182             When `True`, default values are omitted from the response.
2183
2184             Read more about it in the
2185             [FastAPI docs for Response Model - Return Type]
2186             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2187             parameter).
2188             """
2189         ),
2190     ] = False,
2191     response_model_exclude_none: Annotated[
2192         bool,
2193         Doc(
2194             """
2195             Configuration passed to Pydantic to define if the response data should
2196             exclude fields set to `None`.
2197
2198             This is much simpler (less smart) than `response_model_exclude_unset`
2199             and `response_model_exclude_defaults`. You probably want to use one of
2200             those two instead of this one, as those allow returning `None` values
2201             when it makes sense.
2202
2203             Read more about it in the
2204             [FastAPI docs for Response Model - Return Type]
2205             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
2206             """
2207         ),
2208     ] = False,
2209     include_in_schema: Annotated[
2210         bool,
2211         Doc(
2212             """
2213             Include this *path operation* in the generated OpenAPI schema.
2214
2215             This affects the generated OpenAPI (e.g. visible at `/docs`).
2216
2217             Read more about it in the
2218             [FastAPI docs for Query Parameters and String Validations]
2219             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
2220             openapi).
2221             """
2222         ),
2223     ] = True,
2224     response_class: Annotated[
2225         type[Response],
2226         Doc(
2227             """
2228             Response class to be used for this *path operation*.
2229
2230             This will not be used if you return a response directly.
2231
2232             Read more about it in the
2233             [FastAPI docs for Custom Response - HTML, Stream, File, others]
2234             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
2235             """

```

```

2236     ),
2237     ] = Default(JSONResponse),
2238     name: Annotated[
2239         str | None,
2240         Doc(
2241             """
2242             Name for this *path operation*. Only used internally.
2243             """
2244         ),
2245     ] = None,
2246     callbacks: Annotated[
2247         list[BaseRoute] | None,
2248         Doc(
2249             """
2250             List of *path operations* that will be used as OpenAPI callbacks.
2251
2252             This is only for OpenAPI documentation, the callbacks won't be used
2253             directly.
2254
2255             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2256
2257             Read more about it in the
2258             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
2259             callbacks/).
2260             """
2261         ),
2262     ] = None,
2263     openapi_extra: Annotated[
2264         dict[str, Any] | None,
2265         Doc(
2266             """
2267             Extra metadata to be included in the OpenAPI schema for this *path
2268             operation*.
2269
2270             Read more about it in the
2271             [FastAPI docs for Path Operation Advanced Configuration]
2272             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
2273             path-operation-schema).
2274             """
2275         ),
2276     ] = None,
2277     generate_unique_id_function: Annotated[
2278         Callable[[routing.APIRoute], str],
2279         Doc(
2280             """
2281             Customize the function used to generate unique IDs for the *path
2282             operations* shown in the generated OpenAPI.
2283
2284             This is particularly useful when automatically generating clients or
2285             SDKs for your API.
2286
2287             Read more about it in the
2288             [FastAPI docs about how to Generate Clients]
2289             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
2290             """
2291         ),
2292     ] = Default(generate_unique_id),
2293 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
2294     """
2295     Add a *path operation* using an HTTP PUT operation.
2296
2297     ## Example
2298
2299     ```python
2300     from fastapi import FastAPI
2301     from pydantic import BaseModel
2302
2303     class Item(BaseModel):
2304         name: str
2305         description: str | None = None
2306
2307     app = FastAPI()
2308
2309     @app.put("/items/{item_id}")
2310     def replace_item(item_id: str, item: Item):
2311         return {"message": "Item replaced", "id": item_id}

```

```

2311     """
2312     """
    return self.router.put(
        path,
        response_model=response_model,
        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )

```

meth post

```

post(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP POST operation.

EXAMPLE

```

from fastapi import FastAPI
from pydantic import BaseModel

class Item(BaseModel):
    name: str
    description: str | None = None

app = FastAPI()

```

```
@app.post("/items/")
def create_item(item: Item):
    return {"message": "Item created"}
```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: <code>Any</code> DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> <code>None</code> DEFAULT: <code>None</code>
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum]</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends]</code> <code>None</code> DEFAULT: <code>None</code>
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>

PARAMETER	DESCRIPTION
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/applications.py

```

2314 def post(
2315     self,
2316     path: Annotated[
2317         str,
2318         Doc(
2319             """
2320             The URL path to be used for this *path operation*.
2321
2322             For example, in `http://example.com/items`, the path is `/items`.
2323             """
2324         ),
2325     ],
2326     *,
2327     response_model: Annotated[
2328         Any,
2329         Doc(
2330             """
2331             The type to use for the response.
2332
2333             It could be any valid Pydantic *field* type. So, it doesn't have to
2334             be a Pydantic model, it could be other things, like a `list`, `dict`,
2335             etc.
2336
2337             It will be used for:
2338
2339             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
2340               show it as the response (JSON Schema).
2341             * Serialization: you could return an arbitrary object and the
2342               `response_model` would be used to serialize that object into the
2343               corresponding JSON.
2344             * Filtering: the JSON sent to the client will only contain the data
2345               (fields) defined in the `response_model`. If you returned an object
2346               that contains an attribute `password` but the `response_model` does
2347               not include that field, the JSON sent to the client would not have
2348               that `password`.
2349             * Validation: whatever you return will be serialized with the
2350               `response_model`, converting any data as necessary to generate the
2351               corresponding JSON. But if the data in the object returned is not
2352               valid, that would mean a violation of the contract with the client,
2353               so it's an error from the API developer. So, FastAPI will raise an
2354               error and return a 500 error code (Internal Server Error).
2355
2356             Read more about it in the
2357             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
2358             model/).
2359             """
2360         ),
2361     ] = Default(None),
2362     status_code: Annotated[
2363         int | None,
2364         Doc(
2365             """
2366             The default status code to be used for the response.
2367
2368             You could override the status code by returning a response directly.
2369
2370             Read more about it in the
2371             [FastAPI docs for Response Status Code]
2372             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
2373             """
2374         ),
2375     ] = None,
2376     tags: Annotated[
2377         list[str | Enum] | None,
2378         Doc(
2379             """
2380             A list of tags to be applied to the *path operation*.
2381
2382             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2383
2384             Read more about it in the
2385

```

```

2386         [FastAPI docs for Path Operation Configuration]
2387         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
2388         """
2389     ),
2390     ] = None,
2391     dependencies: Annotated[
2392         Sequence[Depends] | None,
2393         Doc(
2394             """
2395             A list of dependencies (using `Depends()`) to be applied to the
2396             *path operation*.
2397
2398             Read more about it in the
2399             [FastAPI docs for Dependencies in path operation decorators]
2400             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
2401             """
2402         ),
2403     ] = None,
2404     summary: Annotated[
2405         str | None,
2406         Doc(
2407             """
2408             A summary for the *path operation*.
2409
2410             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2411
2412             Read more about it in the
2413             [FastAPI docs for Path Operation Configuration]
2414             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
2415             """
2416         ),
2417     ] = None,
2418     description: Annotated[
2419         str | None,
2420         Doc(
2421             """
2422             A description for the *path operation*.
2423
2424             If not provided, it will be extracted automatically from the docstring
2425             of the *path operation function*.
2426
2427             It can contain Markdown.
2428
2429             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2430
2431             Read more about it in the
2432             [FastAPI docs for Path Operation Configuration]
2433             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
2434             """
2435         ),
2436     ] = None,
2437     response_description: Annotated[
2438         str,
2439         Doc(
2440             """
2441             The description for the default response.
2442
2443             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2444             """
2445         ),
2446     ] = "Successful Response",
2447     responses: Annotated[
2448         dict[int | str, dict[str, Any]] | None,
2449         Doc(
2450             """
2451             Additional responses that could be returned by this *path operation*.
2452
2453             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2454             """
2455         ),
2456     ] = None,
2457     deprecated: Annotated[
2458         bool | None,
2459         Doc(
2460             """
2461             Mark this *path operation* as deprecated.

```

```

2462         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2463         """
2464     ),
2465 ] = None,
2466 operation_id: Annotated[
2467     str | None,
2468     Doc(
2469         """
2470         Custom operation ID to be used by this *path operation*.
2471
2472         By default, it is generated automatically.
2473
2474         If you provide a custom operation ID, you need to make sure it is
2475         unique for the whole API.
2476
2477         You can customize the
2478         operation ID generation with the parameter
2479         `generate_unique_id_function` in the `FastAPI` class.
2480
2481         Read more about it in the
2482         [FastAPI docs about how to Generate Clients]
2483         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
2484         """
2485     ),
2486 ] = None,
2487 response_model_include: Annotated[
2488     IncEx | None,
2489     Doc(
2490         """
2491         Configuration passed to Pydantic to include only certain fields in the
2492         response data.
2493
2494         Read more about it in the
2495         [FastAPI docs for Response Model - Return Type]
2496         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2497         """
2498     ),
2499 ] = None,
2500 response_model_exclude: Annotated[
2501     IncEx | None,
2502     Doc(
2503         """
2504         Configuration passed to Pydantic to exclude certain fields in the
2505         response data.
2506
2507         Read more about it in the
2508         [FastAPI docs for Response Model - Return Type]
2509         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2510         """
2511     ),
2512 ] = None,
2513 response_model_by_alias: Annotated[
2514     bool,
2515     Doc(
2516         """
2517         Configuration passed to Pydantic to define if the response model
2518         should be serialized by alias when an alias is used.
2519
2520         Read more about it in the
2521         [FastAPI docs for Response Model - Return Type]
2522         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2523         """
2524     ),
2525 ] = True,
2526 response_model_exclude_unset: Annotated[
2527     bool,
2528     Doc(
2529         """
2530         Configuration passed to Pydantic to define if the response data
2531         should have all the fields, including the ones that were not set and
2532         have their default values. This is different from
2533         `response_model_exclude_defaults` in that if the fields are set,

```

```

2538         they will be included in the response, even if the value is the same
2539         as the default.
2540
2541         When `True`, default values are omitted from the response.
2542
2543         Read more about it in the
2544         [FastAPI docs for Response Model - Return Type]
2545         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2546         parameter).
2547         """
2548     ),
2549     ] = False,
2550     response_model_exclude_defaults: Annotated[
2551         bool,
2552         Doc(
2553             """
2554             Configuration passed to Pydantic to define if the response data
2555             should have all the fields, including the ones that have the same value
2556             as the default. This is different from `response_model_exclude_unset`
2557             in that if the fields are set but contain the same default values,
2558             they will be excluded from the response.
2559
2560             When `True`, default values are omitted from the response.
2561
2562             Read more about it in the
2563             [FastAPI docs for Response Model - Return Type]
2564             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2565             parameter).
2566             """
2567         ),
2568     ] = False,
2569     response_model_exclude_none: Annotated[
2570         bool,
2571         Doc(
2572             """
2573             Configuration passed to Pydantic to define if the response data should
2574             exclude fields set to `None`.
2575
2576             This is much simpler (less smart) than `response_model_exclude_unset`
2577             and `response_model_exclude_defaults`. You probably want to use one of
2578             those two instead of this one, as those allow returning `None` values
2579             when it makes sense.
2580
2581             Read more about it in the
2582             [FastAPI docs for Response Model - Return Type]
2583             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
2584             """
2585         ),
2586     ] = False,
2587     include_in_schema: Annotated[
2588         bool,
2589         Doc(
2590             """
2591             Include this *path operation* in the generated OpenAPI schema.
2592
2593             This affects the generated OpenAPI (e.g. visible at `/docs`).
2594
2595             Read more about it in the
2596             [FastAPI docs for Query Parameters and String Validations]
2597             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
2598             openapi).
2599             """
2600         ),
2601     ] = True,
2602     response_class: Annotated[
2603         type[Response],
2604         Doc(
2605             """
2606             Response class to be used for this *path operation*.
2607
2608             This will not be used if you return a response directly.
2609
2610             Read more about it in the
2611             [FastAPI docs for Custom Response - HTML, Stream, File, others]
2612             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
2613             """

```

```

2614     ),
2615     ] = Default(JSONResponse),
2616     name: Annotated[
2617         str | None,
2618         Doc(
2619             """
2620             Name for this *path operation*. Only used internally.
2621             """
2622         ),
2623     ] = None,
2624     callbacks: Annotated[
2625         list[BaseRoute] | None,
2626         Doc(
2627             """
2628             List of *path operations* that will be used as OpenAPI callbacks.
2629
2630             This is only for OpenAPI documentation, the callbacks won't be used
2631             directly.
2632
2633             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2634
2635             Read more about it in the
2636             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
2637             callbacks/).
2638             """
2639         ),
2640     ] = None,
2641     openapi_extra: Annotated[
2642         dict[str, Any] | None,
2643         Doc(
2644             """
2645             Extra metadata to be included in the OpenAPI schema for this *path
2646             operation*.
2647
2648             Read more about it in the
2649             [FastAPI docs for Path Operation Advanced Configuration]
2650             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
2651             path-operation-schema).
2652             """
2653         ),
2654     ] = None,
2655     generate_unique_id_function: Annotated[
2656         Callable[[routing.APIRoute], str],
2657         Doc(
2658             """
2659             Customize the function used to generate unique IDs for the *path
2660             operations* shown in the generated OpenAPI.
2661
2662             This is particularly useful when automatically generating clients or
2663             SDKs for your API.
2664
2665             Read more about it in the
2666             [FastAPI docs about how to Generate Clients]
2667             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
2668             """
2669         ),
2670     ] = Default(generate_unique_id),
2671 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
2672     """
2673     Add a *path operation* using an HTTP POST operation.
2674
2675     ## Example
2676
2677     ```python
2678     from fastapi import FastAPI
2679     from pydantic import BaseModel
2680
2681     class Item(BaseModel):
2682         name: str
2683         description: str | None = None
2684
2685     app = FastAPI()
2686
2687     @app.post("/items/")
2688     def create_item(item: Item):
2689         return {"message": "Item created"}

```



```

2689     """
2690     """
    return self.router.post(
        path,
        response_model=response_model,
        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )

```

meth delete

```

delete(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP DELETE operation.

EXAMPLE

```

from fastapi import FastAPI

app = FastAPI()

@app.delete("/items/{item_id}")
def delete_item(item_id: str):
    return {"message": "Item deleted"}

```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: Any DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> None DEFAULT: None
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum]</code> None DEFAULT: None
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends]</code> None DEFAULT: None

PARAMETER	DESCRIPTION
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>

PARAMETER	DESCRIPTION
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/applications.py

```

2692 def delete(
2693     self,
2694     path: Annotated[
2695         str,
2696         Doc(
2697             """
2698             The URL path to be used for this *path operation*.
2699
2700             For example, in `http://example.com/items`, the path is `/items`.
2701             """
2702         ),
2703     ],
2704     *,
2705     response_model: Annotated[
2706         Any,
2707         Doc(
2708             """
2709             The type to use for the response.
2710
2711             It could be any valid Pydantic *field* type. So, it doesn't have to
2712             be a Pydantic model, it could be other things, like a `list`, `dict`,
2713             etc.
2714
2715             It will be used for:
2716
2717             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
2718               show it as the response (JSON Schema).
2719             * Serialization: you could return an arbitrary object and the
2720               `response_model` would be used to serialize that object into the
2721               corresponding JSON.
2722             * Filtering: the JSON sent to the client will only contain the data
2723               (fields) defined in the `response_model`. If you returned an object
2724               that contains an attribute `password` but the `response_model` does
2725               not include that field, the JSON sent to the client would not have
2726               that `password`.
2727             * Validation: whatever you return will be serialized with the
2728               `response_model`, converting any data as necessary to generate the
2729               corresponding JSON. But if the data in the object returned is not
2730               valid, that would mean a violation of the contract with the client,
2731               so it's an error from the API developer. So, FastAPI will raise an
2732               error and return a 500 error code (Internal Server Error).
2733
2734             Read more about it in the
2735             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
2736             model/).
2737             """
2738         ),
2739     ] = Default(None),
2740     status_code: Annotated[
2741         int | None,
2742         Doc(
2743             """
2744             The default status code to be used for the response.
2745
2746             You could override the status code by returning a response directly.
2747
2748             Read more about it in the
2749             [FastAPI docs for Response Status Code]
2750             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
2751             """
2752         ),
2753     ] = None,
2754     tags: Annotated[
2755         list[str | Enum] | None,
2756         Doc(
2757             """
2758             A list of tags to be applied to the *path operation*.
2759
2760             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2761
2762             Read more about it in the
2763

```

```

2764         [FastAPI docs for Path Operation Configuration]
2765         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
2766         """
2767     ),
2768     ] = None,
2769     dependencies: Annotated[
2770         Sequence[Depends] | None,
2771         Doc(
2772             """
2773             A list of dependencies (using `Depends()`) to be applied to the
2774             *path operation*.
2775
2776             Read more about it in the
2777             [FastAPI docs for Dependencies in path operation decorators]
2778             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
2779             """
2780         ),
2781     ] = None,
2782     summary: Annotated[
2783         str | None,
2784         Doc(
2785             """
2786             A summary for the *path operation*.
2787
2788             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2789
2790             Read more about it in the
2791             [FastAPI docs for Path Operation Configuration]
2792             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
2793             """
2794         ),
2795     ] = None,
2796     description: Annotated[
2797         str | None,
2798         Doc(
2799             """
2800             A description for the *path operation*.
2801
2802             If not provided, it will be extracted automatically from the docstring
2803             of the *path operation function*.
2804
2805             It can contain Markdown.
2806
2807             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2808
2809             Read more about it in the
2810             [FastAPI docs for Path Operation Configuration]
2811             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
2812             """
2813         ),
2814     ] = None,
2815     response_description: Annotated[
2816         str,
2817         Doc(
2818             """
2819             The description for the default response.
2820
2821             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2822             """
2823         ),
2824     ] = "Successful Response",
2825     responses: Annotated[
2826         dict[int | str, dict[str, Any]] | None,
2827         Doc(
2828             """
2829             Additional responses that could be returned by this *path operation*.
2830
2831             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2832             """
2833         ),
2834     ] = None,
2835     deprecated: Annotated[
2836         bool | None,
2837         Doc(
2838             """
2839             Mark this *path operation* as deprecated.

```

```

2840         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
2841         """
2842     ),
2843 ] = None,
2844 operation_id: Annotated[
2845     str | None,
2846     Doc(
2847         """
2848         Custom operation ID to be used by this *path operation*.
2849
2850         By default, it is generated automatically.
2851
2852         If you provide a custom operation ID, you need to make sure it is
2853         unique for the whole API.
2854
2855         You can customize the
2856         operation ID generation with the parameter
2857         `generate_unique_id_function` in the `FastAPI` class.
2858
2859         Read more about it in the
2860         [FastAPI docs about how to Generate Clients]
2861         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
2862         """
2863     ),
2864 ] = None,
2865 response_model_include: Annotated[
2866     IncEx | None,
2867     Doc(
2868         """
2869         Configuration passed to Pydantic to include only certain fields in the
2870         response data.
2871
2872         Read more about it in the
2873         [FastAPI docs for Response Model - Return Type]
2874         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2875         """
2876     ),
2877 ] = None,
2878 response_model_exclude: Annotated[
2879     IncEx | None,
2880     Doc(
2881         """
2882         Configuration passed to Pydantic to exclude certain fields in the
2883         response data.
2884
2885         Read more about it in the
2886         [FastAPI docs for Response Model - Return Type]
2887         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2888         """
2889     ),
2890 ] = None,
2891 response_model_by_alias: Annotated[
2892     bool,
2893     Doc(
2894         """
2895         Configuration passed to Pydantic to define if the response model
2896         should be serialized by alias when an alias is used.
2897
2898         Read more about it in the
2899         [FastAPI docs for Response Model - Return Type]
2900         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
2901         """
2902     ),
2903 ] = True,
2904 response_model_exclude_unset: Annotated[
2905     bool,
2906     Doc(
2907         """
2908         Configuration passed to Pydantic to define if the response data
2909         should have all the fields, including the ones that were not set and
2910         have their default values. This is different from
2911         `response_model_exclude_defaults` in that if the fields are set,

```



```

2916         they will be included in the response, even if the value is the same
2917         as the default.
2918
2919         When `True`, default values are omitted from the response.
2920
2921         Read more about it in the
2922         [FastAPI docs for Response Model - Return Type]
2923         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2924         parameter).
2925         """
2926     ),
2927     ] = False,
2928     response_model_exclude_defaults: Annotated[
2929         bool,
2930         Doc(
2931             """
2932             Configuration passed to Pydantic to define if the response data
2933             should have all the fields, including the ones that have the same value
2934             as the default. This is different from `response_model_exclude_unset`
2935             in that if the fields are set but contain the same default values,
2936             they will be excluded from the response.
2937
2938             When `True`, default values are omitted from the response.
2939
2940             Read more about it in the
2941             [FastAPI docs for Response Model - Return Type]
2942             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
2943             parameter).
2944             """
2945         ),
2946     ] = False,
2947     response_model_exclude_none: Annotated[
2948         bool,
2949         Doc(
2950             """
2951             Configuration passed to Pydantic to define if the response data should
2952             exclude fields set to `None`.
2953
2954             This is much simpler (less smart) than `response_model_exclude_unset`
2955             and `response_model_exclude_defaults`. You probably want to use one of
2956             those two instead of this one, as those allow returning `None` values
2957             when it makes sense.
2958
2959             Read more about it in the
2960             [FastAPI docs for Response Model - Return Type]
2961             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
2962             """
2963         ),
2964     ] = False,
2965     include_in_schema: Annotated[
2966         bool,
2967         Doc(
2968             """
2969             Include this *path operation* in the generated OpenAPI schema.
2970
2971             This affects the generated OpenAPI (e.g. visible at `/docs`).
2972
2973             Read more about it in the
2974             [FastAPI docs for Query Parameters and String Validations]
2975             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
2976             openapi).
2977             """
2978         ),
2979     ] = True,
2980     response_class: Annotated[
2981         type[Response],
2982         Doc(
2983             """
2984             Response class to be used for this *path operation*.
2985
2986             This will not be used if you return a response directly.
2987
2988             Read more about it in the
2989             [FastAPI docs for Custom Response - HTML, Stream, File, others]
2990             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
2991             """

```

```

2992     ),
2993     ] = Default(JSONResponse),
2994     name: Annotated[
2995         str | None,
2996         Doc(
2997             """
2998             Name for this *path operation*. Only used internally.
2999             """
3000         ),
3001     ] = None,
3002     callbacks: Annotated[
3003         list[BaseRoute] | None,
3004         Doc(
3005             """
3006             List of *path operations* that will be used as OpenAPI callbacks.
3007
3008             This is only for OpenAPI documentation, the callbacks won't be used
3009             directly.
3010
3011             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3012
3013             Read more about it in the
3014             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
3015             callbacks/).
3016             """
3017         ),
3018     ] = None,
3019     openapi_extra: Annotated[
3020         dict[str, Any] | None,
3021         Doc(
3022             """
3023             Extra metadata to be included in the OpenAPI schema for this *path
3024             operation*.
3025
3026             Read more about it in the
3027             [FastAPI docs for Path Operation Advanced Configuration]
3028             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
3029             path-operation-schema).
3030             """
3031         ),
3032     ] = None,
3033     generate_unique_id_function: Annotated[
3034         Callable[[routing.APIRoute], str],
3035         Doc(
3036             """
3037             Customize the function used to generate unique IDs for the *path
3038             operations* shown in the generated OpenAPI.
3039
3040             This is particularly useful when automatically generating clients or
3041             SDKs for your API.
3042
3043             Read more about it in the
3044             [FastAPI docs about how to Generate Clients]
3045             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
3046             """
3047         ),
3048     ] = Default(generate_unique_id),
3049 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
3050     """
3051     Add a *path operation* using an HTTP DELETE operation.
3052
3053     ## Example
3054
3055     ```python
3056     from fastapi import FastAPI
3057
3058     app = FastAPI()
3059
3060     @app.delete("/items/{item_id}")
3061     def delete_item(item_id: str):
3062         return {"message": "Item deleted"}
3063     """
3064     return self.router.delete(
3065         path,
3066         response_model=response_model,

```

```

        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )

```

meth options

```

options(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP OPTIONS operation.

EXAMPLE

```

from fastapi import FastAPI

app = FastAPI()

@app.options("/items/")
def get_item_options():
    return {"additions": ["Aji", "Guacamole"]}

```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: Any DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> None DEFAULT: None
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum]</code> None DEFAULT: None
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends]</code> None DEFAULT: None

PARAMETER	DESCRIPTION
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>

PARAMETER	DESCRIPTION
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/applications.py

```

3065 def options(
3066     self,
3067     path: Annotated[
3068         str,
3069         Doc(
3070             """
3071             The URL path to be used for this *path operation*.
3072
3073             For example, in `http://example.com/items`, the path is `/items`.
3074             """
3075         ),
3076     ],
3077     *,
3078     response_model: Annotated[
3079         Any,
3080         Doc(
3081             """
3082             The type to use for the response.
3083
3084             It could be any valid Pydantic *field* type. So, it doesn't have to
3085             be a Pydantic model, it could be other things, like a `list`, `dict`,
3086             etc.
3087
3088             It will be used for:
3089
3090             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
3091               show it as the response (JSON Schema).
3092             * Serialization: you could return an arbitrary object and the
3093               `response_model` would be used to serialize that object into the
3094               corresponding JSON.
3095             * Filtering: the JSON sent to the client will only contain the data
3096               (fields) defined in the `response_model`. If you returned an object
3097               that contains an attribute `password` but the `response_model` does
3098               not include that field, the JSON sent to the client would not have
3099               that `password`.
3100             * Validation: whatever you return will be serialized with the
3101               `response_model`, converting any data as necessary to generate the
3102               corresponding JSON. But if the data in the object returned is not
3103               valid, that would mean a violation of the contract with the client,
3104               so it's an error from the API developer. So, FastAPI will raise an
3105               error and return a 500 error code (Internal Server Error).
3106
3107             Read more about it in the
3108             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
3109             model/).
3110             """
3111         ),
3112     ] = Default(None),
3113     status_code: Annotated[
3114         int | None,
3115         Doc(
3116             """
3117             The default status code to be used for the response.
3118
3119             You could override the status code by returning a response directly.
3120
3121             Read more about it in the
3122             [FastAPI docs for Response Status Code]
3123             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
3124             """
3125         ),
3126     ] = None,
3127     tags: Annotated[
3128         list[str | Enum] | None,
3129         Doc(
3130             """
3131             A list of tags to be applied to the *path operation*.
3132
3133             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3134
3135             Read more about it in the
3136

```



```

3137         [FastAPI docs for Path Operation Configuration]
3138         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
3139         """
3140     ),
3141     ] = None,
3142     dependencies: Annotated[
3143         Sequence[Depends] | None,
3144         Doc(
3145             """
3146             A list of dependencies (using `Depends()`) to be applied to the
3147             *path operation*.
3148
3149             Read more about it in the
3150             [FastAPI docs for Dependencies in path operation decorators]
3151             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
3152             """
3153         ),
3154     ] = None,
3155     summary: Annotated[
3156         str | None,
3157         Doc(
3158             """
3159             A summary for the *path operation*.
3160
3161             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3162
3163             Read more about it in the
3164             [FastAPI docs for Path Operation Configuration]
3165             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3166             """
3167         ),
3168     ] = None,
3169     description: Annotated[
3170         str | None,
3171         Doc(
3172             """
3173             A description for the *path operation*.
3174
3175             If not provided, it will be extracted automatically from the docstring
3176             of the *path operation function*.
3177
3178             It can contain Markdown.
3179
3180             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3181
3182             Read more about it in the
3183             [FastAPI docs for Path Operation Configuration]
3184             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3185             """
3186         ),
3187     ] = None,
3188     response_description: Annotated[
3189         str,
3190         Doc(
3191             """
3192             The description for the default response.
3193
3194             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3195             """
3196         ),
3197     ] = "Successful Response",
3198     responses: Annotated[
3199         dict[int | str, dict[str, Any]] | None,
3200         Doc(
3201             """
3202             Additional responses that could be returned by this *path operation*.
3203
3204             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3205             """
3206         ),
3207     ] = None,
3208     deprecated: Annotated[
3209         bool | None,
3210         Doc(
3211             """
3212             Mark this *path operation* as deprecated.

```

```

3213         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3214         """
3215     ),
3216     ] = None,
3217     operation_id: Annotated[
3218         str | None,
3219         Doc(
3220             """
3221             Custom operation ID to be used by this *path operation*.
3222
3223             By default, it is generated automatically.
3224
3225             If you provide a custom operation ID, you need to make sure it is
3226             unique for the whole API.
3227
3228             You can customize the
3229             operation ID generation with the parameter
3230             `generate_unique_id_function` in the `FastAPI` class.
3231
3232             Read more about it in the
3233             [FastAPI docs about how to Generate Clients]
3234             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
3235             """
3236         ),
3237     ] = None,
3238     response_model_include: Annotated[
3239         IncEx | None,
3240         Doc(
3241             """
3242             Configuration passed to Pydantic to include only certain fields in the
3243             response data.
3244
3245             Read more about it in the
3246             [FastAPI docs for Response Model - Return Type]
3247             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3248             """
3249         ),
3250     ] = None,
3251     response_model_exclude: Annotated[
3252         IncEx | None,
3253         Doc(
3254             """
3255             Configuration passed to Pydantic to exclude certain fields in the
3256             response data.
3257
3258             Read more about it in the
3259             [FastAPI docs for Response Model - Return Type]
3260             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3261             """
3262         ),
3263     ] = None,
3264     response_model_by_alias: Annotated[
3265         bool,
3266         Doc(
3267             """
3268             Configuration passed to Pydantic to define if the response model
3269             should be serialized by alias when an alias is used.
3270
3271             Read more about it in the
3272             [FastAPI docs for Response Model - Return Type]
3273             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3274             """
3275         ),
3276     ] = True,
3277     response_model_exclude_unset: Annotated[
3278         bool,
3279         Doc(
3280             """
3281             Configuration passed to Pydantic to define if the response data
3282             should have all the fields, including the ones that were not set and
3283             have their default values. This is different from
3284             `response_model_exclude_defaults` in that if the fields are set,

```

```

3289         they will be included in the response, even if the value is the same
3290         as the default.
3291
3292         When `True`, default values are omitted from the response.
3293
3294         Read more about it in the
3295         [FastAPI docs for Response Model - Return Type]
3296         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
3297         parameter).
3298         """
3299     ),
3300     ] = False,
3301     response_model_exclude_defaults: Annotated[
3302         bool,
3303         Doc(
3304             """
3305             Configuration passed to Pydantic to define if the response data
3306             should have all the fields, including the ones that have the same value
3307             as the default. This is different from `response_model_exclude_unset`
3308             in that if the fields are set but contain the same default values,
3309             they will be excluded from the response.
3310
3311             When `True`, default values are omitted from the response.
3312
3313             Read more about it in the
3314             [FastAPI docs for Response Model - Return Type]
3315             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
3316             parameter).
3317             """
3318         ),
3319     ] = False,
3320     response_model_exclude_none: Annotated[
3321         bool,
3322         Doc(
3323             """
3324             Configuration passed to Pydantic to define if the response data should
3325             exclude fields set to `None`.
3326
3327             This is much simpler (less smart) than `response_model_exclude_unset`
3328             and `response_model_exclude_defaults`. You probably want to use one of
3329             those two instead of this one, as those allow returning `None` values
3330             when it makes sense.
3331
3332             Read more about it in the
3333             [FastAPI docs for Response Model - Return Type]
3334             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
3335             """
3336         ),
3337     ] = False,
3338     include_in_schema: Annotated[
3339         bool,
3340         Doc(
3341             """
3342             Include this *path operation* in the generated OpenAPI schema.
3343
3344             This affects the generated OpenAPI (e.g. visible at `/docs`).
3345
3346             Read more about it in the
3347             [FastAPI docs for Query Parameters and String Validations]
3348             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
3349             openapi).
3350             """
3351         ),
3352     ] = True,
3353     response_class: Annotated[
3354         type[Response],
3355         Doc(
3356             """
3357             Response class to be used for this *path operation*.
3358
3359             This will not be used if you return a response directly.
3360
3361             Read more about it in the
3362             [FastAPI docs for Custom Response - HTML, Stream, File, others]
3363             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
3364             """

```

```

3365     ),
3366     ] = Default(JSONResponse),
3367     name: Annotated[
3368         str | None,
3369         Doc(
3370             """
3371             Name for this *path operation*. Only used internally.
3372             """
3373         ),
3374     ] = None,
3375     callbacks: Annotated[
3376         list[BaseRoute] | None,
3377         Doc(
3378             """
3379             List of *path operations* that will be used as OpenAPI callbacks.
3380
3381             This is only for OpenAPI documentation, the callbacks won't be used
3382             directly.
3383
3384             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3385
3386             Read more about it in the
3387             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
3388             callbacks/).
3389             """
3390         ),
3391     ] = None,
3392     openapi_extra: Annotated[
3393         dict[str, Any] | None,
3394         Doc(
3395             """
3396             Extra metadata to be included in the OpenAPI schema for this *path
3397             operation*.
3398
3399             Read more about it in the
3400             [FastAPI docs for Path Operation Advanced Configuration]
3401             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
3402             path-operation-schema).
3403             """
3404         ),
3405     ] = None,
3406     generate_unique_id_function: Annotated[
3407         Callable[[routing.APIRoute], str],
3408         Doc(
3409             """
3410             Customize the function used to generate unique IDs for the *path
3411             operations* shown in the generated OpenAPI.
3412
3413             This is particularly useful when automatically generating clients or
3414             SDKs for your API.
3415
3416             Read more about it in the
3417             [FastAPI docs about how to Generate Clients]
3418             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
3419             """
3420         ),
3421     ] = Default(generate_unique_id),
3422 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
3423     """
3424     Add a *path operation* using an HTTP OPTIONS operation.
3425
3426     ## Example
3427
3428     ```python
3429     from fastapi import FastAPI
3430
3431     app = FastAPI()
3432
3433     @app.options("/items/")
3434     def get_item_options():
3435         return {"additions": ["Aji", "Guacamole"]}
3436     """
3437     return self.router.options(
3438         path,
3439         response_model=response_model,

```

```

        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )

```

meth head

```

def head(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP HEAD operation.

EXAMPLE

```

from fastapi import FastAPI, Response

app = FastAPI()

@app.head("/items/", status_code=204)
def get_items_headers(response: Response):
    response.headers["X-Cat-Dog"] = "Alone in the world"

```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: Any DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> None DEFAULT: None
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum]</code> None DEFAULT: None
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends]</code> None DEFAULT: None

PARAMETER	DESCRIPTION
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>

PARAMETER	DESCRIPTION
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/applications.py

```

3438 def head(
3439     self,
3440     path: Annotated[
3441         str,
3442         Doc(
3443             """
3444             The URL path to be used for this *path operation*.
3445
3446             For example, in `http://example.com/items`, the path is `/items`.
3447             """
3448         ),
3449     ],
3450     *,
3451     response_model: Annotated[
3452         Any,
3453         Doc(
3454             """
3455             The type to use for the response.
3456
3457             It could be any valid Pydantic *field* type. So, it doesn't have to
3458             be a Pydantic model, it could be other things, like a `list`, `dict`,
3459             etc.
3460
3461             It will be used for:
3462
3463             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
3464               show it as the response (JSON Schema).
3465             * Serialization: you could return an arbitrary object and the
3466               `response_model` would be used to serialize that object into the
3467               corresponding JSON.
3468             * Filtering: the JSON sent to the client will only contain the data
3469               (fields) defined in the `response_model`. If you returned an object
3470               that contains an attribute `password` but the `response_model` does
3471               not include that field, the JSON sent to the client would not have
3472               that `password`.
3473             * Validation: whatever you return will be serialized with the
3474               `response_model`, converting any data as necessary to generate the
3475               corresponding JSON. But if the data in the object returned is not
3476               valid, that would mean a violation of the contract with the client,
3477               so it's an error from the API developer. So, FastAPI will raise an
3478               error and return a 500 error code (Internal Server Error).
3479
3480             Read more about it in the
3481             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
3482             model/).
3483             """
3484         ),
3485     ] = Default(None),
3486     status_code: Annotated[
3487         int | None,
3488         Doc(
3489             """
3490             The default status code to be used for the response.
3491
3492             You could override the status code by returning a response directly.
3493
3494             Read more about it in the
3495             [FastAPI docs for Response Status Code]
3496             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
3497             """
3498         ),
3499     ] = None,
3500     tags: Annotated[
3501         list[str | Enum] | None,
3502         Doc(
3503             """
3504             A list of tags to be applied to the *path operation*.
3505
3506             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3507
3508             Read more about it in the
3509

```

```

3510         [FastAPI docs for Path Operation Configuration]
3511         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
3512         """
3513     ),
3514     ] = None,
3515     dependencies: Annotated[
3516         Sequence[Depends] | None,
3517         Doc(
3518             """
3519             A list of dependencies (using `Depends()`) to be applied to the
3520             *path operation*.
3521
3522             Read more about it in the
3523             [FastAPI docs for Dependencies in path operation decorators]
3524             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
3525             """
3526         ),
3527     ] = None,
3528     summary: Annotated[
3529         str | None,
3530         Doc(
3531             """
3532             A summary for the *path operation*.
3533
3534             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3535
3536             Read more about it in the
3537             [FastAPI docs for Path Operation Configuration]
3538             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3539             """
3540         ),
3541     ] = None,
3542     description: Annotated[
3543         str | None,
3544         Doc(
3545             """
3546             A description for the *path operation*.
3547
3548             If not provided, it will be extracted automatically from the docstring
3549             of the *path operation* function*.
3550
3551             It can contain Markdown.
3552
3553             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3554
3555             Read more about it in the
3556             [FastAPI docs for Path Operation Configuration]
3557             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3558             """
3559         ),
3560     ] = None,
3561     response_description: Annotated[
3562         str,
3563         Doc(
3564             """
3565             The description for the default response.
3566
3567             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3568             """
3569         ),
3570     ] = "Successful Response",
3571     responses: Annotated[
3572         dict[int | str, dict[str, Any]] | None,
3573         Doc(
3574             """
3575             Additional responses that could be returned by this *path operation*.
3576
3577             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3578             """
3579         ),
3580     ] = None,
3581     deprecated: Annotated[
3582         bool | None,
3583         Doc(
3584             """
3585             Mark this *path operation* as deprecated.

```

```

3586         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3587         """
3588     ),
3589 ] = None,
3590 operation_id: Annotated[
3591     str | None,
3592     Doc(
3593         """
3594         Custom operation ID to be used by this *path operation*.
3595
3596         By default, it is generated automatically.
3597
3598         If you provide a custom operation ID, you need to make sure it is
3599         unique for the whole API.
3600
3601         You can customize the
3602         operation ID generation with the parameter
3603         `generate_unique_id_function` in the `FastAPI` class.
3604
3605         Read more about it in the
3606         [FastAPI docs about how to Generate Clients]
3607         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
3608         """
3609     ),
3610 ] = None,
3611 response_model_include: Annotated[
3612     IncEx | None,
3613     Doc(
3614         """
3615         Configuration passed to Pydantic to include only certain fields in the
3616         response data.
3617
3618         Read more about it in the
3619         [FastAPI docs for Response Model - Return Type]
3620         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3621         """
3622     ),
3623 ] = None,
3624 response_model_exclude: Annotated[
3625     IncEx | None,
3626     Doc(
3627         """
3628         Configuration passed to Pydantic to exclude certain fields in the
3629         response data.
3630
3631         Read more about it in the
3632         [FastAPI docs for Response Model - Return Type]
3633         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3634         """
3635     ),
3636 ] = None,
3637 response_model_by_alias: Annotated[
3638     bool,
3639     Doc(
3640         """
3641         Configuration passed to Pydantic to define if the response model
3642         should be serialized by alias when an alias is used.
3643
3644         Read more about it in the
3645         [FastAPI docs for Response Model - Return Type]
3646         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3647         """
3648     ),
3649 ] = True,
3650 response_model_exclude_unset: Annotated[
3651     bool,
3652     Doc(
3653         """
3654         Configuration passed to Pydantic to define if the response data
3655         should have all the fields, including the ones that were not set and
3656         have their default values. This is different from
3657         `response_model_exclude_defaults` in that if the fields are set,
3658

```

```

3662         they will be included in the response, even if the value is the same
3663         as the default.
3664
3665         When `True`, default values are omitted from the response.
3666
3667         Read more about it in the
3668         [FastAPI docs for Response Model - Return Type]
3669         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
3670         parameter).
3671         """
3672     ),
3673     ] = False,
3674     response_model_exclude_defaults: Annotated[
3675         bool,
3676         Doc(
3677             """
3678             Configuration passed to Pydantic to define if the response data
3679             should have all the fields, including the ones that have the same value
3680             as the default. This is different from `response_model_exclude_unset`
3681             in that if the fields are set but contain the same default values,
3682             they will be excluded from the response.
3683
3684             When `True`, default values are omitted from the response.
3685
3686             Read more about it in the
3687             [FastAPI docs for Response Model - Return Type]
3688             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
3689             parameter).
3690             """
3691         ),
3692     ] = False,
3693     response_model_exclude_none: Annotated[
3694         bool,
3695         Doc(
3696             """
3697             Configuration passed to Pydantic to define if the response data should
3698             exclude fields set to `None`.
3699
3700             This is much simpler (less smart) than `response_model_exclude_unset`
3701             and `response_model_exclude_defaults`. You probably want to use one of
3702             those two instead of this one, as those allow returning `None` values
3703             when it makes sense.
3704
3705             Read more about it in the
3706             [FastAPI docs for Response Model - Return Type]
3707             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
3708             """
3709         ),
3710     ] = False,
3711     include_in_schema: Annotated[
3712         bool,
3713         Doc(
3714             """
3715             Include this *path operation* in the generated OpenAPI schema.
3716
3717             This affects the generated OpenAPI (e.g. visible at `/docs`).
3718
3719             Read more about it in the
3720             [FastAPI docs for Query Parameters and String Validations]
3721             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
3722             openapi).
3723             """
3724         ),
3725     ] = True,
3726     response_class: Annotated[
3727         type[Response],
3728         Doc(
3729             """
3730             Response class to be used for this *path operation*.
3731
3732             This will not be used if you return a response directly.
3733
3734             Read more about it in the
3735             [FastAPI docs for Custom Response - HTML, Stream, File, others]
3736             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
3737             """

```

```

3738     ),
3739     ] = Default(JSONResponse),
3740     name: Annotated[
3741         str | None,
3742         Doc(
3743             """
3744             Name for this *path operation*. Only used internally.
3745             """
3746         ),
3747     ] = None,
3748     callbacks: Annotated[
3749         list[BaseRoute] | None,
3750         Doc(
3751             """
3752             List of *path operations* that will be used as OpenAPI callbacks.
3753
3754             This is only for OpenAPI documentation, the callbacks won't be used
3755             directly.
3756
3757             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3758
3759             Read more about it in the
3760             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
3761 callbacks/).
3762             """
3763         ),
3764     ] = None,
3765     openapi_extra: Annotated[
3766         dict[str, Any] | None,
3767         Doc(
3768             """
3769             Extra metadata to be included in the OpenAPI schema for this *path
3770             operation*.
3771
3772             Read more about it in the
3773             [FastAPI docs for Path Operation Advanced Configuration]
3774 (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
3775 path-operation-schema).
3776             """
3777         ),
3778     ] = None,
3779     generate_unique_id_function: Annotated[
3780         Callable[[routing.APIRoute], str],
3781         Doc(
3782             """
3783             Customize the function used to generate unique IDs for the *path
3784             operations* shown in the generated OpenAPI.
3785
3786             This is particularly useful when automatically generating clients or
3787             SDKs for your API.
3788
3789             Read more about it in the
3790             [FastAPI docs about how to Generate Clients]
3791 (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
3792             """
3793         ),
3794     ] = Default(generate_unique_id),
3795 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
3796     """
3797     Add a *path operation* using an HTTP HEAD operation.
3798
3799     ## Example
3800
3801     ```python
3802     from fastapi import FastAPI, Response
3803
3804     app = FastAPI()
3805
3806     @app.head("/items/", status_code=204)
3807     def get_items_headers(response: Response):
3808         response.headers["X-Cat-Dog"] = "Alone in the world"
3809     """
3809     return self.router.head(
3810         path,
3811         response_model=response_model,

```

```

        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )

```

meth patch

```

patch(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP PATCH operation.

EXAMPLE

```

from fastapi import FastAPI
from pydantic import BaseModel

class Item(BaseModel):
    name: str
    description: str | None = None

app = FastAPI()

@app.patch("/items/")
def update_item(item: Item):
    return {"message": "Item updated in place"}

```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: Any DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> <code>None</code> DEFAULT: <code>None</code>
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum]</code> <code>None</code> DEFAULT: <code>None</code>
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends]</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>

PARAMETER	DESCRIPTION
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/applications.py

```

3811 def patch(
3812     self,
3813     path: Annotated[
3814         str,
3815         Doc(
3816             """
3817             The URL path to be used for this *path operation*.
3818
3819             For example, in `http://example.com/items`, the path is `/items`.
3820             """
3821         ),
3822     ],
3823     *,
3824     response_model: Annotated[
3825         Any,
3826         Doc(
3827             """
3828             The type to use for the response.
3829
3830             It could be any valid Pydantic *field* type. So, it doesn't have to
3831             be a Pydantic model, it could be other things, like a `list`, `dict`,
3832             etc.
3833
3834             It will be used for:
3835
3836             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
3837               show it as the response (JSON Schema).
3838             * Serialization: you could return an arbitrary object and the
3839               `response_model` would be used to serialize that object into the
3840               corresponding JSON.
3841             * Filtering: the JSON sent to the client will only contain the data
3842               (fields) defined in the `response_model`. If you returned an object
3843               that contains an attribute `password` but the `response_model` does
3844               not include that field, the JSON sent to the client would not have
3845               that `password`.
3846             * Validation: whatever you return will be serialized with the
3847               `response_model`, converting any data as necessary to generate the
3848               corresponding JSON. But if the data in the object returned is not
3849               valid, that would mean a violation of the contract with the client,
3850               so it's an error from the API developer. So, FastAPI will raise an
3851               error and return a 500 error code (Internal Server Error).
3852
3853             Read more about it in the
3854             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
3855             model/).
3856             """
3857         ),
3858     ] = Default(None),
3859     status_code: Annotated[
3860         int | None,
3861         Doc(
3862             """
3863             The default status code to be used for the response.
3864
3865             You could override the status code by returning a response directly.
3866
3867             Read more about it in the
3868             [FastAPI docs for Response Status Code]
3869             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
3870             """
3871         ),
3872     ] = None,
3873     tags: Annotated[
3874         list[str | Enum] | None,
3875         Doc(
3876             """
3877             A list of tags to be applied to the *path operation*.
3878
3879             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3880
3881             Read more about it in the
3882

```

```

3883         [FastAPI docs for Path Operation Configuration]
3884         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
3885         """
3886     ),
3887     ] = None,
3888     dependencies: Annotated[
3889         Sequence[Depends] | None,
3890         Doc(
3891             """
3892             A list of dependencies (using `Depends()`) to be applied to the
3893             *path operation*.
3894
3895             Read more about it in the
3896             [FastAPI docs for Dependencies in path operation decorators]
3897             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
3898             """
3899         ),
3900     ] = None,
3901     summary: Annotated[
3902         str | None,
3903         Doc(
3904             """
3905             A summary for the *path operation*.
3906
3907             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3908
3909             Read more about it in the
3910             [FastAPI docs for Path Operation Configuration]
3911             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3912             """
3913         ),
3914     ] = None,
3915     description: Annotated[
3916         str | None,
3917         Doc(
3918             """
3919             A description for the *path operation*.
3920
3921             If not provided, it will be extracted automatically from the docstring
3922             of the *path operation function*.
3923
3924             It can contain Markdown.
3925
3926             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3927
3928             Read more about it in the
3929             [FastAPI docs for Path Operation Configuration]
3930             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
3931             """
3932         ),
3933     ] = None,
3934     response_description: Annotated[
3935         str,
3936         Doc(
3937             """
3938             The description for the default response.
3939
3940             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3941             """
3942         ),
3943     ] = "Successful Response",
3944     responses: Annotated[
3945         dict[int | str, dict[str, Any]] | None,
3946         Doc(
3947             """
3948             Additional responses that could be returned by this *path operation*.
3949
3950             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3951             """
3952         ),
3953     ] = None,
3954     deprecated: Annotated[
3955         bool | None,
3956         Doc(
3957             """
3958             Mark this *path operation* as deprecated.

```

```

3959         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
3960         """
3961     ),
3962 ] = None,
3963 operation_id: Annotated[
3964     str | None,
3965     Doc(
3966         """
3967         Custom operation ID to be used by this *path operation*.
3968
3969         By default, it is generated automatically.
3970
3971         If you provide a custom operation ID, you need to make sure it is
3972         unique for the whole API.
3973
3974         You can customize the
3975         operation ID generation with the parameter
3976         `generate_unique_id_function` in the `FastAPI` class.
3977
3978         Read more about it in the
3979         [FastAPI docs about how to Generate Clients]
3980         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
3981         """
3982     ),
3983 ] = None,
3984 response_model_include: Annotated[
3985     IncEx | None,
3986     Doc(
3987         """
3988         Configuration passed to Pydantic to include only certain fields in the
3989         response data.
3990
3991         Read more about it in the
3992         [FastAPI docs for Response Model - Return Type]
3993         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
3994         """
3995     ),
3996 ] = None,
3997 response_model_exclude: Annotated[
3998     IncEx | None,
3999     Doc(
4000         """
4001         Configuration passed to Pydantic to exclude certain fields in the
4002         response data.
4003
4004         Read more about it in the
4005         [FastAPI docs for Response Model - Return Type]
4006         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
4007         """
4008     ),
4009 ] = None,
4010 response_model_by_alias: Annotated[
4011     bool,
4012     Doc(
4013         """
4014         Configuration passed to Pydantic to define if the response model
4015         should be serialized by alias when an alias is used.
4016
4017         Read more about it in the
4018         [FastAPI docs for Response Model - Return Type]
4019         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
4020         """
4021     ),
4022 ] = True,
4023 response_model_exclude_unset: Annotated[
4024     bool,
4025     Doc(
4026         """
4027         Configuration passed to Pydantic to define if the response data
4028         should have all the fields, including the ones that were not set and
4029         have their default values. This is different from
4030         `response_model_exclude_defaults` in that if the fields are set,
4031

```

```

4035         they will be included in the response, even if the value is the same
4036         as the default.
4037
4038         When `True`, default values are omitted from the response.
4039
4040         Read more about it in the
4041         [FastAPI docs for Response Model - Return Type]
4042         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
4043         parameter).
4044         """
4045     ),
4046     ] = False,
4047     response_model_exclude_defaults: Annotated[
4048         bool,
4049         Doc(
4050             """
4051             Configuration passed to Pydantic to define if the response data
4052             should have all the fields, including the ones that have the same value
4053             as the default. This is different from `response_model_exclude_unset`
4054             in that if the fields are set but contain the same default values,
4055             they will be excluded from the response.
4056
4057             When `True`, default values are omitted from the response.
4058
4059             Read more about it in the
4060             [FastAPI docs for Response Model - Return Type]
4061             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
4062             parameter).
4063             """
4064         ),
4065     ] = False,
4066     response_model_exclude_none: Annotated[
4067         bool,
4068         Doc(
4069             """
4070             Configuration passed to Pydantic to define if the response data should
4071             exclude fields set to `None`.
4072
4073             This is much simpler (less smart) than `response_model_exclude_unset`
4074             and `response_model_exclude_defaults`. You probably want to use one of
4075             those two instead of this one, as those allow returning `None` values
4076             when it makes sense.
4077
4078             Read more about it in the
4079             [FastAPI docs for Response Model - Return Type]
4080             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
4081             """
4082         ),
4083     ] = False,
4084     include_in_schema: Annotated[
4085         bool,
4086         Doc(
4087             """
4088             Include this *path operation* in the generated OpenAPI schema.
4089
4090             This affects the generated OpenAPI (e.g. visible at `/docs`).
4091
4092             Read more about it in the
4093             [FastAPI docs for Query Parameters and String Validations]
4094             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
4095             openapi).
4096             """
4097         ),
4098     ] = True,
4099     response_class: Annotated[
4100         type[Response],
4101         Doc(
4102             """
4103             Response class to be used for this *path operation*.
4104
4105             This will not be used if you return a response directly.
4106
4107             Read more about it in the
4108             [FastAPI docs for Custom Response - HTML, Stream, File, others]
4109             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
4110             """

```

```

4111     ),
4112     ] = Default(JSONResponse),
4113     name: Annotated[
4114         str | None,
4115         Doc(
4116             """
4117             Name for this *path operation*. Only used internally.
4118             """
4119         ),
4120     ] = None,
4121     callbacks: Annotated[
4122         list[BaseRoute] | None,
4123         Doc(
4124             """
4125             List of *path operations* that will be used as OpenAPI callbacks.
4126
4127             This is only for OpenAPI documentation, the callbacks won't be used
4128             directly.
4129
4130             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4131
4132             Read more about it in the
4133             [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
4134             callbacks/).
4135             """
4136         ),
4137     ] = None,
4138     openapi_extra: Annotated[
4139         dict[str, Any] | None,
4140         Doc(
4141             """
4142             Extra metadata to be included in the OpenAPI schema for this *path
4143             operation*.
4144
4145             Read more about it in the
4146             [FastAPI docs for Path Operation Advanced Configuration]
4147             (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
4148             path-operation-schema).
4149             """
4150         ),
4151     ] = None,
4152     generate_unique_id_function: Annotated[
4153         Callable[[routing.APIRoute], str],
4154         Doc(
4155             """
4156             Customize the function used to generate unique IDs for the *path
4157             operations* shown in the generated OpenAPI.
4158
4159             This is particularly useful when automatically generating clients or
4160             SDKs for your API.
4161
4162             Read more about it in the
4163             [FastAPI docs about how to Generate Clients]
4164             (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
4165             """
4166         ),
4167     ] = Default(generate_unique_id),
4168 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
4169     """
4170     Add a *path operation* using an HTTP PATCH operation.
4171
4172     ## Example
4173
4174     ```python
4175     from fastapi import FastAPI
4176     from pydantic import BaseModel
4177
4178     class Item(BaseModel):
4179         name: str
4180         description: str | None = None
4181
4182     app = FastAPI()
4183
4184     @app.patch("/items/")
4185     def update_item(item: Item):
4186         return {"message": "Item updated in place"}

```



```

4186     """
4187     """
    return self.router.patch(
        path,
        response_model=response_model,
        status_code=status_code,
        tags=tags,
        dependencies=dependencies,
        summary=summary,
        description=description,
        response_description=response_description,
        responses=responses,
        deprecated=deprecated,
        operation_id=operation_id,
        response_model_include=response_model_include,
        response_model_exclude=response_model_exclude,
        response_model_by_alias=response_model_by_alias,
        response_model_exclude_unset=response_model_exclude_unset,
        response_model_exclude_defaults=response_model_exclude_defaults,
        response_model_exclude_none=response_model_exclude_none,
        include_in_schema=include_in_schema,
        response_class=response_class,
        name=name,
        callbacks=callbacks,
        openapi_extra=openapi_extra,
        generate_unique_id_function=generate_unique_id_function,
    )

```

meth trace

```

trace(
    path,
    *,
    response_model=Default(None),
    status_code=None,
    tags=None,
    dependencies=None,
    summary=None,
    description=None,
    response_description="Successful Response",
    responses=None,
    deprecated=None,
    operation_id=None,
    response_model_include=None,
    response_model_exclude=None,
    response_model_by_alias=True,
    response_model_exclude_unset=False,
    response_model_exclude_defaults=False,
    response_model_exclude_none=False,
    include_in_schema=True,
    response_class=Default(JSONResponse),
    name=None,
    callbacks=None,
    openapi_extra=None,
    generate_unique_id_function=Default(generate_unique_id)
)

```

Add a *path operation* using an HTTP TRACE operation.

EXAMPLE

```

from fastapi import FastAPI

app = FastAPI()

@app.trace("/items/{item_id}")
def trace_item(item_id: str):
    return None

```

PARAMETER	DESCRIPTION
<code>path</code>	The URL path to be used for this <i>path operation</i>. For example, in <code>http://example.com/items</code>, the path is <code>/items</code>. TYPE: <code>str</code>
<code>response_model</code>	The type to use for the response. It could be any valid Pydantic <i>field</i> type. So, it doesn't have to be a Pydantic model, it could be other things, like a <code>list</code>, <code>dict</code>, etc. It will be used for: • Documentation: the generated OpenAPI (and the UI at <code>/docs</code>) will show it as the response (JSON Schema). • Serialization: you could return an arbitrary object and the <code>response_model</code> would be used to serialize that object into the corresponding JSON. • Filtering: the JSON sent to the client will only contain the data (fields) defined in the <code>response_model</code>. If you returned an object that contains an attribute <code>password</code> but the <code>response_model</code> does not include that field, the JSON sent to the client would not have that <code>password</code>. • Validation: whatever you return will be serialized with the <code>response_model</code>, converting any data as necessary to generate the corresponding JSON. But if the data in the object returned is not valid, that would mean a violation of the contract with the client, so it's an error from the API developer. So, FastAPI will raise an error and return a <code>500</code> error code (Internal Server Error). Read more about it in the FastAPI docs for Response Model. TYPE: Any DEFAULT: <code>Default(None)</code>
<code>status_code</code>	The default status code to be used for the response. You could override the status code by returning a response directly. Read more about it in the FastAPI docs for Response Status Code. TYPE: <code>int</code> None DEFAULT: None
<code>tags</code>	A list of tags to be applied to the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>list[str Enum]</code> None DEFAULT: None
<code>dependencies</code>	A list of dependencies (using <code>Depends()</code>) to be applied to the <i>path operation</i>. Read more about it in the FastAPI docs for Dependencies in path operation decorators. TYPE: <code>Sequence[Depends]</code> None DEFAULT: None

PARAMETER	DESCRIPTION
<code>summary</code>	A summary for the <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>description</code>	A description for the <i>path operation</i>. If not provided, it will be extracted automatically from the docstring of the <i>path operation function</i>. It can contain Markdown. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Path Operation Configuration. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_description</code>	The description for the default response. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>str</code> DEFAULT: <code>'Successful Response'</code>
<code>responses</code>	Additional responses that could be returned by this <i>path operation</i>. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>dict[int str, dict[str, Any]]</code> <code>None</code> DEFAULT: <code>None</code>
<code>deprecated</code>	Mark this <i>path operation</i> as deprecated. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). TYPE: <code>bool</code> <code>None</code> DEFAULT: <code>None</code>
<code>operation_id</code>	Custom operation ID to be used by this <i>path operation</i>. By default, it is generated automatically. If you provide a custom operation ID, you need to make sure it is unique for the whole API. You can customize the operation ID generation with the parameter <code>generate_unique_id_function</code> in the <code>FastAPI</code> class. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>str</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_include</code>	Configuration passed to Pydantic to include only certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>

PARAMETER	DESCRIPTION
<code>response_model_exclude</code>	Configuration passed to Pydantic to exclude certain fields in the response data. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>IncEx</code> <code>None</code> DEFAULT: <code>None</code>
<code>response_model_by_alias</code>	Configuration passed to Pydantic to define if the response model should be serialized by alias when an alias is used. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>True</code>
<code>response_model_exclude_unset</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that were not set and have their default values. This is different from <code>response_model_exclude_defaults</code> in that if the fields are set, they will be included in the response, even if the value is the same as the default. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_defaults</code>	Configuration passed to Pydantic to define if the response data should have all the fields, including the ones that have the same value as the default. This is different from <code>response_model_exclude_unset</code> in that if the fields are set but contain the same default values, they will be excluded from the response. When <code>True</code>, default values are omitted from the response. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>response_model_exclude_none</code>	Configuration passed to Pydantic to define if the response data should exclude fields set to <code>None</code>. This is much simpler (less smart) than <code>response_model_exclude_unset</code> and <code>response_model_exclude_defaults</code>. You probably want to use one of those two instead of this one, as those allow returning <code>None</code> values when it makes sense. Read more about it in the FastAPI docs for Response Model - Return Type. TYPE: <code>bool</code> DEFAULT: <code>False</code>
<code>include_in_schema</code>	Include this <i>path operation</i> in the generated OpenAPI schema. This affects the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for Query Parameters and String Validations. TYPE: <code>bool</code> DEFAULT: <code>True</code>

PARAMETER	DESCRIPTION
<code>response_class</code>	Response class to be used for this <i>path operation</i>. This will not be used if you return a response directly. Read more about it in the FastAPI docs for Custom Response - HTML, Stream, File, others. TYPE: <code>type[Response]</code> DEFAULT: <code>Default(JSONResponse)</code>
<code>name</code>	Name for this <i>path operation</i>. Only used internally. TYPE: <code>str None</code> DEFAULT: <code>None</code>
<code>callbacks</code>	List of <i>path operations</i> that will be used as OpenAPI callbacks. This is only for OpenAPI documentation, the callbacks won't be used directly. It will be added to the generated OpenAPI (e.g. visible at <code>/docs</code>). Read more about it in the FastAPI docs for OpenAPI Callbacks. TYPE: <code>list[BaseRoute] None</code> DEFAULT: <code>None</code>
<code>openapi_extra</code>	Extra metadata to be included in the OpenAPI schema for this <i>path operation</i>. Read more about it in the FastAPI docs for Path Operation Advanced Configuration. TYPE: <code>dict[str, Any] None</code> DEFAULT: <code>None</code>
<code>generate_unique_id_function</code>	Customize the function used to generate unique IDs for the <i>path operations</i> shown in the generated OpenAPI. This is particularly useful when automatically generating clients or SDKs for your API. Read more about it in the FastAPI docs about how to Generate Clients. TYPE: <code>Callable[[APIRoute], str]</code> DEFAULT: <code>Default(generate_unique_id)</code>

<> Source code in fastapi/applications.py

```

4189 def trace(
4190     self,
4191     path: Annotated[
4192         str,
4193         Doc(
4194             """
4195             The URL path to be used for this *path operation*.
4196
4197             For example, in `http://example.com/items`, the path is `/items`.
4198             """
4199         ),
4200     ],
4201     *,
4202     response_model: Annotated[
4203         Any,
4204         Doc(
4205             """
4206             The type to use for the response.
4207
4208             It could be any valid Pydantic *field* type. So, it doesn't have to
4209             be a Pydantic model, it could be other things, like a `list`, `dict`,
4210             etc.
4211
4212             It will be used for:
4213
4214             * Documentation: the generated OpenAPI (and the UI at `/docs`) will
4215               show it as the response (JSON Schema).
4216             * Serialization: you could return an arbitrary object and the
4217               `response_model` would be used to serialize that object into the
4218               corresponding JSON.
4219             * Filtering: the JSON sent to the client will only contain the data
4220               (fields) defined in the `response_model`. If you returned an object
4221               that contains an attribute `password` but the `response_model` does
4222               not include that field, the JSON sent to the client would not have
4223               that `password`.
4224             * Validation: whatever you return will be serialized with the
4225               `response_model`, converting any data as necessary to generate the
4226               corresponding JSON. But if the data in the object returned is not
4227               valid, that would mean a violation of the contract with the client,
4228               so it's an error from the API developer. So, FastAPI will raise an
4229               error and return a 500 error code (Internal Server Error).
4230
4231             Read more about it in the
4232             [FastAPI docs for Response Model](https://fastapi.tiangolo.com/tutorial/response-
4233             model/).
4234             """
4235         ),
4236     ] = Default(None),
4237     status_code: Annotated[
4238         int | None,
4239         Doc(
4240             """
4241             The default status code to be used for the response.
4242
4243             You could override the status code by returning a response directly.
4244
4245             Read more about it in the
4246             [FastAPI docs for Response Status Code]
4247             (https://fastapi.tiangolo.com/tutorial/response-status-code/).
4248             """
4249         ),
4250     ] = None,
4251     tags: Annotated[
4252         list[str | Enum] | None,
4253         Doc(
4254             """
4255             A list of tags to be applied to the *path operation*.
4256
4257             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4258
4259             Read more about it in the
4260

```

```

4261         [FastAPI docs for Path Operation Configuration]
4262         (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/#tags).
4263         """
4264     ),
4265     ] = None,
4266     dependencies: Annotated[
4267         Sequence[Depends] | None,
4268         Doc(
4269             """
4270             A list of dependencies (using `Depends()`) to be applied to the
4271             *path operation*.
4272
4273             Read more about it in the
4274             [FastAPI docs for Dependencies in path operation decorators]
4275             (https://fastapi.tiangolo.com/tutorial/dependencies/dependencies-in-path-operation-decorators/).
4276             """
4277         ),
4278     ] = None,
4279     summary: Annotated[
4280         str | None,
4281         Doc(
4282             """
4283             A summary for the *path operation*.
4284
4285             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4286
4287             Read more about it in the
4288             [FastAPI docs for Path Operation Configuration]
4289             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
4290             """
4291         ),
4292     ] = None,
4293     description: Annotated[
4294         str | None,
4295         Doc(
4296             """
4297             A description for the *path operation*.
4298
4299             If not provided, it will be extracted automatically from the docstring
4300             of the *path operation function*.
4301
4302             It can contain Markdown.
4303
4304             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4305
4306             Read more about it in the
4307             [FastAPI docs for Path Operation Configuration]
4308             (https://fastapi.tiangolo.com/tutorial/path-operation-configuration/).
4309             """
4310         ),
4311     ] = None,
4312     response_description: Annotated[
4313         str,
4314         Doc(
4315             """
4316             The description for the default response.
4317
4318             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4319             """
4320         ),
4321     ] = "Successful Response",
4322     responses: Annotated[
4323         dict[int | str, dict[str, Any]] | None,
4324         Doc(
4325             """
4326             Additional responses that could be returned by this *path operation*.
4327
4328             It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4329             """
4330         ),
4331     ] = None,
4332     deprecated: Annotated[
4333         bool | None,
4334         Doc(
4335             """
4336             Mark this *path operation* as deprecated.

```

```

4337         It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4338         """
4339     ),
4340 ] = None,
4341 operation_id: Annotated[
4342     str | None,
4343     Doc(
4344         """
4345         Custom operation ID to be used by this *path operation*.
4346
4347         By default, it is generated automatically.
4348
4349         If you provide a custom operation ID, you need to make sure it is
4350         unique for the whole API.
4351
4352         You can customize the
4353         operation ID generation with the parameter
4354         `generate_unique_id_function` in the `FastAPI` class.
4355
4356         Read more about it in the
4357         [FastAPI docs about how to Generate Clients]
4358         (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
4359         """
4360     ),
4361 ] = None,
4362 response_model_include: Annotated[
4363     IncEx | None,
4364     Doc(
4365         """
4366         Configuration passed to Pydantic to include only certain fields in the
4367         response data.
4368
4369         Read more about it in the
4370         [FastAPI docs for Response Model - Return Type]
4371         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
4372         """
4373     ),
4374 ] = None,
4375 response_model_exclude: Annotated[
4376     IncEx | None,
4377     Doc(
4378         """
4379         Configuration passed to Pydantic to exclude certain fields in the
4380         response data.
4381
4382         Read more about it in the
4383         [FastAPI docs for Response Model - Return Type]
4384         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
4385         """
4386     ),
4387 ] = None,
4388 response_model_by_alias: Annotated[
4389     bool,
4390     Doc(
4391         """
4392         Configuration passed to Pydantic to define if the response model
4393         should be serialized by alias when an alias is used.
4394
4395         Read more about it in the
4396         [FastAPI docs for Response Model - Return Type]
4397         (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_include-and-response\_model\_exclude).
4398         """
4399     ),
4400 ] = True,
4401 response_model_exclude_unset: Annotated[
4402     bool,
4403     Doc(
4404         """
4405         Configuration passed to Pydantic to define if the response data
4406         should have all the fields, including the ones that were not set and
4407         have their default values. This is different from
4408         `response_model_exclude_defaults` in that if the fields are set,

```



```

4413         they will be included in the response, even if the value is the same
4414         as the default.
4415
4416         When `True`, default values are omitted from the response.
4417
4418         Read more about it in the
4419         [FastAPI docs for Response Model - Return Type]
4420         (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
4421         parameter).
4422         """
4423     ),
4424     ] = False,
4425     response_model_exclude_defaults: Annotated[
4426         bool,
4427         Doc(
4428             """
4429             Configuration passed to Pydantic to define if the response data
4430             should have all the fields, including the ones that have the same value
4431             as the default. This is different from `response_model_exclude_unset`
4432             in that if the fields are set but contain the same default values,
4433             they will be excluded from the response.
4434
4435             When `True`, default values are omitted from the response.
4436
4437             Read more about it in the
4438             [FastAPI docs for Response Model - Return Type]
4439             (https://fastapi.tiangolo.com/tutorial/response-model/#use-the-response\_model\_exclude\_unset-
4440             parameter).
4441             """
4442         ),
4443     ] = False,
4444     response_model_exclude_none: Annotated[
4445         bool,
4446         Doc(
4447             """
4448             Configuration passed to Pydantic to define if the response data should
4449             exclude fields set to `None`.
4450
4451             This is much simpler (less smart) than `response_model_exclude_unset`
4452             and `response_model_exclude_defaults`. You probably want to use one of
4453             those two instead of this one, as those allow returning `None` values
4454             when it makes sense.
4455
4456             Read more about it in the
4457             [FastAPI docs for Response Model - Return Type]
4458             (https://fastapi.tiangolo.com/tutorial/response-model/#response\_model\_exclude\_none).
4459             """
4460         ),
4461     ] = False,
4462     include_in_schema: Annotated[
4463         bool,
4464         Doc(
4465             """
4466             Include this *path operation* in the generated OpenAPI schema.
4467
4468             This affects the generated OpenAPI (e.g. visible at `/docs`).
4469
4470             Read more about it in the
4471             [FastAPI docs for Query Parameters and String Validations]
4472             (https://fastapi.tiangolo.com/tutorial/query-params-str-validations/#exclude-parameters-from-
4473             openapi).
4474             """
4475         ),
4476     ] = True,
4477     response_class: Annotated[
4478         type[Response],
4479         Doc(
4480             """
4481             Response class to be used for this *path operation*.
4482
4483             This will not be used if you return a response directly.
4484
4485             Read more about it in the
4486             [FastAPI docs for Custom Response - HTML, Stream, File, others]
4487             (https://fastapi.tiangolo.com/advanced/custom-response/#redirectresponse).
4488             """

```

```

4489         ),
4490         ] = Default(JSONResponse),
4491         name: Annotated[
4492             str | None,
4493             Doc(
4494                 """
4495                 Name for this *path operation*. Only used internally.
4496                 """
4497             ),
4498         ] = None,
4499         callbacks: Annotated[
4500             list[BaseRoute] | None,
4501             Doc(
4502                 """
4503                 List of *path operations* that will be used as OpenAPI callbacks.
4504
4505                 This is only for OpenAPI documentation, the callbacks won't be used
4506                 directly.
4507
4508                 It will be added to the generated OpenAPI (e.g. visible at `/docs`).
4509
4510                 Read more about it in the
4511                 [FastAPI docs for OpenAPI Callbacks](https://fastapi.tiangolo.com/advanced/openapi-
4512                 callbacks/).
4513                 """
4514             ),
4515         ] = None,
4516         openapi_extra: Annotated[
4517             dict[str, Any] | None,
4518             Doc(
4519                 """
4520                 Extra metadata to be included in the OpenAPI schema for this *path
4521                 operation*.
4522
4523                 Read more about it in the
4524                 [FastAPI docs for Path Operation Advanced Configuration]
4525                 (https://fastapi.tiangolo.com/advanced/path-operation-advanced-configuration/#custom-openapi-
4526                 path-operation-schema).
4527                 """
4528             ),
4529         ] = None,
4530         generate_unique_id_function: Annotated[
4531             Callable[[routing.APIRoute], str],
4532             Doc(
4533                 """
4534                 Customize the function used to generate unique IDs for the *path
4535                 operations* shown in the generated OpenAPI.
4536
4537                 This is particularly useful when automatically generating clients or
4538                 SDKs for your API.
4539
4540                 Read more about it in the
4541                 [FastAPI docs about how to Generate Clients]
4542                 (https://fastapi.tiangolo.com/advanced/generate-clients/#custom-generate-unique-id-function).
4543                 """
4544             ),
4545         ] = Default(generate_unique_id),
4546     ) -> Callable[[DecoratedCallable], DecoratedCallable]:
4547         """
4548         Add a *path operation* using an HTTP TRACE operation.
4549
4550         ## Example
4551
4552         ```python
4553         from fastapi import FastAPI
4554
4555         app = FastAPI()
4556
4557         @app.trace("/items/{item_id}")
4558         def trace_item(item_id: str):
4559             return None
4560         """
4561         return self.router.trace(
4562             path,
4563             response_model=response_model,

```

```
status_code=status_code,
tags=tags,
dependencies=dependencies,
summary=summary,
description=description,
response_description=response_description,
responses=responses,
deprecated=deprecated,
operation_id=operation_id,
response_model_include=response_model_include,
response_model_exclude=response_model_exclude,
response_model_by_alias=response_model_by_alias,
response_model_exclude_unset=response_model_exclude_unset,
response_model_exclude_defaults=response_model_exclude_defaults,
response_model_exclude_none=response_model_exclude_none,
include_in_schema=include_in_schema,
response_class=response_class,
name=name,
callbacks=callbacks,
openapi_extra=openapi_extra,
generate_unique_id_function=generate_unique_id_function,
)
```

meth on_event

```
on_event(event_type)
```

Add an event handler for the application.

`on_event` is deprecated, use `lifespan` event handlers instead.

Read more about it in the [FastAPI docs for Lifespan Events](#).

PARAMETER	DESCRIPTION
<code>event_type</code>	The type of event. <code>startup</code> or <code>shutdown</code> . TYPE: <code>str</code>

<> Source code in `fastapi/applications.py`

```

4571 @deprecated(
4572     """
4573     on_event is deprecated, use lifespan event handlers instead.
4574
4575     Read more about it in the
4576     [FastAPI docs for Lifespan Events](https://fastapi.tiangolo.com/advanced/events/).
4577     """
4578 )
4579 def on_event(
4580     self,
4581     event_type: Annotated[
4582         str,
4583         Doc(
4584             """
4585             The type of event. `startup` or `shutdown`.
4586             """
4587         ),
4588     ],
4589 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
4590     """
4591     Add an event handler for the application.
4592
4593     `on_event` is deprecated, use `lifespan` event handlers instead.
4594
4595     Read more about it in the
4596     [FastAPI docs for Lifespan Events](https://fastapi.tiangolo.com/advanced/events/#alternative-
4597     events-deprecated).
4598     """
4599     return self.router.on_event(event_type) # ty: ignore[deprecated]

```

meth middleware

```
middleware(middleware_type)
```



Add a middleware to the application.

Read more about it in the [FastAPI docs for Middleware](https://fastapi.tiangolo.com/advanced/middleware/).

EXAMPLE

```

import time
from typing import Awaitable, Callable

from fastapi import FastAPI, Request, Response

app = FastAPI()

@app.middleware("http")
async def add_process_time_header(
    request: Request, call_next: Callable[[Request], Awaitable[Response]]
) -> Response:
    start_time = time.time()
    response = await call_next(request)
    process_time = time.time() - start_time
    response.headers["X-Process-Time"] = str(process_time)
    return response

```



PARAMETER	DESCRIPTION
<code>middleware_type</code>	The type of middleware. Currently only supports <code>http</code> .
	TYPE: <code>str</code>

< > Source code in `fastapi/applications.py`

```

4600 def middleware(
4601     self,
4602     middleware_type: Annotated[
4603         str,
4604         Doc(
4605             """
4606             The type of middleware. Currently only supports `http`.
4607             """
4608         ),
4609     ],
4610 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
4611     """
4612     Add a middleware to the application.
4613
4614     Read more about it in the
4615     [FastAPI docs for Middleware](https://fastapi.tiangolo.com/tutorial/middleware/).
4616
4617     ## Example
4618
4619     ```python
4620     import time
4621     from typing import Awaitable, Callable
4622
4623     from fastapi import FastAPI, Request, Response
4624
4625     app = FastAPI()
4626
4627     @app.middleware("http")
4628     async def add_process_time_header(
4629         request: Request, call_next: Callable[[Request], Awaitable[Response]]
4630     ) -> Response:
4631         start_time = time.time()
4632         response = await call_next(request)
4633         process_time = time.time() - start_time
4634         response.headers["X-Process-Time"] = str(process_time)
4635         return response
4636     ```
4637
4638     """
4639
4640     def decorator(func: DecoratedCallable) -> DecoratedCallable:
4641         self.add_middleware(BaseHTTPMiddleware, dispatch=func) # ty: ignore[invalid-argument-
4642     type]
4643         return func
4644
4645     return decorator

```

`meth` exception_handler

`exception_handler(exc_class_or_status_code)`

Add an exception handler to the app.

Read more about it in the [FastAPI docs for Handling Errors](https://fastapi.tiangolo.com/tutorial/handling-errors/).

EXAMPLE

```
from fastapi import FastAPI, Request
from fastapi.responses import JSONResponse

class UnicornException(Exception):
    def __init__(self, name: str):
        self.name = name

app = FastAPI()

@app.exception_handler(UnicornException)
async def unicorn_exception_handler(request: Request, exc: UnicornException):
    return JSONResponse(
        status_code=418,
        content={"message": f"Oops! {exc.name} did something. There goes a rainbow..."}
    )
```

PARAMETER	DESCRIPTION
<code>exc_class_or_status_code</code>	The Exception class this would handle, or a status code. TYPE: <code>int</code> <code>type[Exception]</code>

<> Source code in `fastapi/applications.py`

```

4646 def exception_handler(
4647     self,
4648     exc_class_or_status_code: Annotated[
4649         int | type[Exception],
4650         Doc(
4651             """
4652             The Exception class this would handle, or a status code.
4653             """
4654         ),
4655     ],
4656 ) -> Callable[[DecoratedCallable], DecoratedCallable]:
4657     """
4658     Add an exception handler to the app.
4659
4660     Read more about it in the
4661     [FastAPI docs for Handling Errors](https://fastapi.tiangolo.com/tutorial/handling-errors/).
4662
4663     ## Example
4664
4665     ```python
4666     from fastapi import FastAPI, Request
4667     from fastapi.responses import JSONResponse
4668
4669     class UnicornException(Exception):
4670         def __init__(self, name: str):
4671             self.name = name
4672
4673     app = FastAPI()
4674
4675     @app.exception_handler(UnicornException)
4676     async def unicorn_exception_handler(request: Request, exc: UnicornException):
4677         return JSONResponse(
4678             status_code=418,
4679             content={"message": f"Oops! {exc.name} did something. There goes a rainbow..."},
4680         )
4681     ```
4682     """
4683
4684     def decorator(func: DecoratedCallable) -> DecoratedCallable:
4685         self.add_exception_handler(exc_class_or_status_code, func)
4686         return func
4687
4688     return decorator
4689
4690
4691

```